



General Physics II

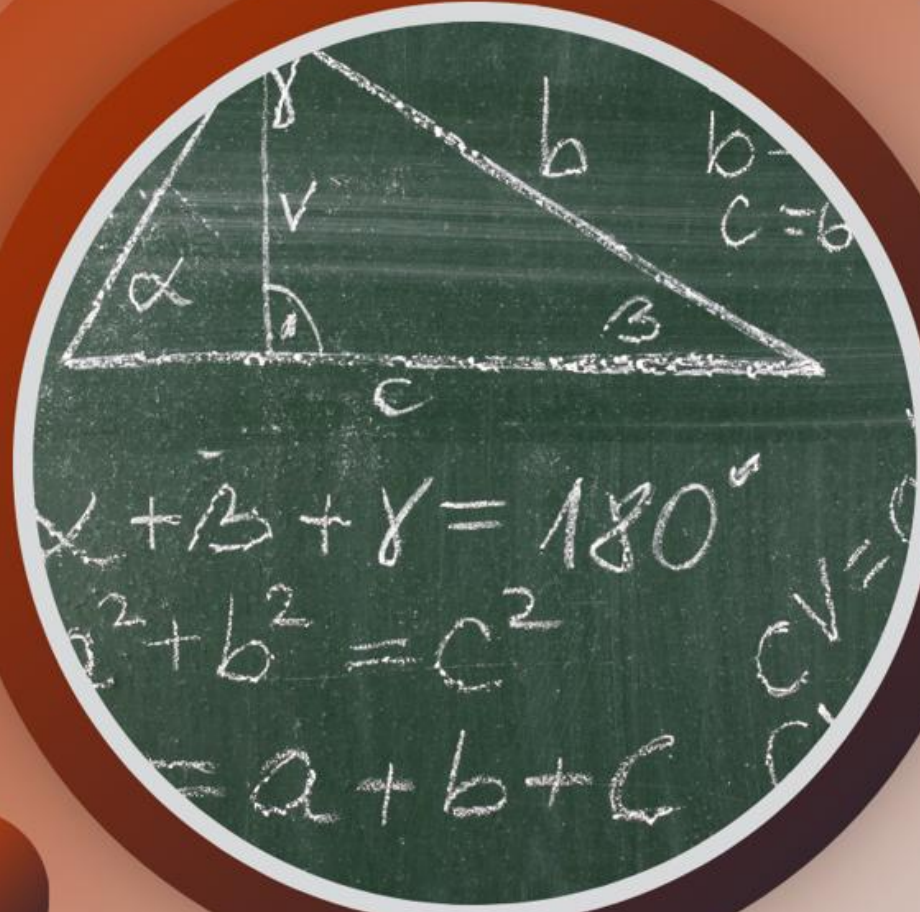
PHY121

College of Science and Technology



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MODULE 3: ELECTRICITY & CIRCUIT ANALYSIS

UNIT 1: Electrical Currents, Resistance, and semiconductors

Electric Current, Resistance, and Semiconductors

- All electric circuits have one thing in common—they depend on the flow of electric charge.
- When electric charge flows from one place to another, we say it forms an **electric current**. The more charge that flows, and the faster it flows, the greater the electric current.
- In general, electric charge is carried through a circuit by electrons.

Electric Current, Resistance, and Semiconductors

- Suppose an amount of charge ΔQ flows past a given point in a wire in the time Δt . The electric current, I , in the wire is simply defined as the amount of charge divided by the amount of time.
- The following equation is used to determine the current flowing in a wire.

Definition of Electric Current, I

electric current = $\frac{\text{amount of charge}}{\text{amount of time}}$

$$I = \frac{\Delta Q}{\Delta t}$$

SI unit: coulombs per second (C/s) = ampere (A)

Electric Current, Resistance, and Semiconductors

- The unit of current is the *ampere* (A), or *amp* for short. It is named for the French physicist André-Marie Ampère (1775–1836).
- A current of 1 amp is defined as the flow of 1 coulomb of charge in 1 second:
$$1 \text{ A} = 1 \text{ C/s}$$
- A 1-amp current is fairly strong. Many electronic devices, like cell phones and digital music players, operate on currents that are a fraction of an amp.

Electric Current, Resistance, and Semiconductors

- The following Conceptual Example illustrates how the current depends on both the amount of charge flowing and the amount of time.

CONCEPTUAL Example 21.1 Comparing Currents

Which of the following situations has the greatest electric current?

Which has the smallest electric current?

- (a) the flow of 6 C in 3 s
- (b) the flow of 3 C in 1 s
- (c) a person walking 8 m in 4 s

Reasoning and Discussion

The electric current is charge per time. Therefore, the current in situation

(a) is $I = \frac{6 \text{ C}}{3 \text{ s}} = 2 \text{ A}$, and the current in situation (b) is $I = \frac{3 \text{ C}}{1 \text{ s}} = 3 \text{ A}$.

The current in situation (c) is zero because a person's total charge is essentially zero. As a result, when a person moves from one place to another, there is practically no *net* flow of electric charge.

Answer

Situation (b) has the greatest current, 3 A; situation (c) has the smallest current, zero.

Electric Current, Resistance, and Semiconductors

- The following example shows that the number of electrons flowing in a typical circuit is extremely large. The situation is similar to the large number of water molecules flowing through a garden hose.

GUIDED Example 21.2 | Mega Blaster

Electric Current

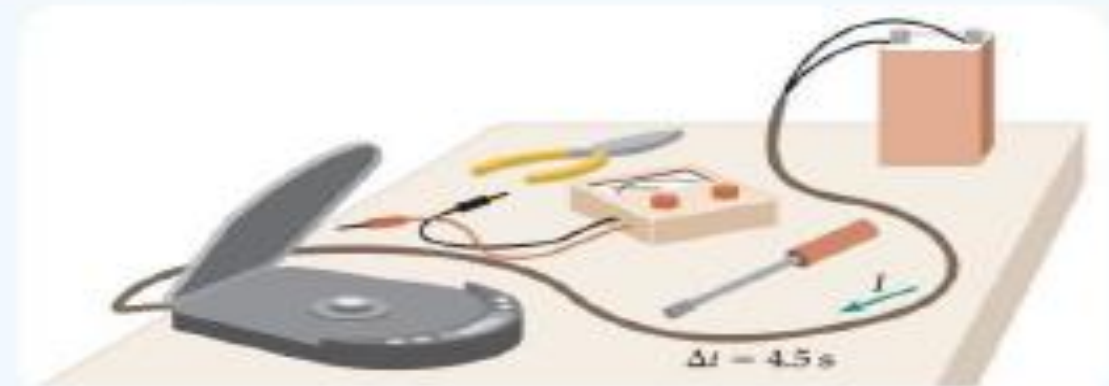
The disk drive in a portable CD player is connected to a battery that supplies it with a current of 0.22 A . How many electrons pass through the disk drive in 4.5 s ? (Recall that the charge of an electron has the magnitude $e = 1.60 \times 10^{-19} \text{ C}$.)

Picture the Problem

Our sketch shows the CD player with a current $I = 0.22 \text{ A}$ flowing through it. Also indicated is the time, $\Delta t = 4.5 \text{ s}$, during which the current flows.

Strategy

Since we know both the current, I , and the amount of time, Δt , we can use the definition of current, $I = \Delta Q / \Delta t$, to find the amount of charge, ΔQ , that flows through the player. Once we know the amount of charge, the number of electrons, N , is that amount divided by the magnitude of the electron's charge. That is, $N = \Delta Q / e$, where $e = 1.60 \times 10^{-19} \text{ C}$.



Known

$$I = 0.22 \text{ A}$$

$$\Delta t = 4.5 \text{ s}$$

$$e = 1.60 \times 10^{-19} \text{ C}$$

Unknown

$$\Delta Q = ?$$

$$N = ?$$

Electric Current, Resistance, and Semiconductors

GUIDED Example 21.2 | Mega Blaster (Continued)

Electric Current

Solution

- 1 Rearrange $I = \Delta Q / \Delta t$ to solve for the amount of charge, ΔQ , that flows through the drive:

$$\begin{aligned}\Delta Q &= I \Delta t \\ &= (0.22 \text{ A})(4.5 \text{ s}) \\ &= 0.99 \text{ C}\end{aligned}$$

Math HELP
Significant
Figures
See Lesson 1.4

- 2 Divide the amount of charge, ΔQ , by the magnitude of the electron's charge, e , to find the number of electrons, N :

$$\begin{aligned}N &= \frac{\Delta Q}{e} \\ &= \frac{0.99 \text{ C}}{1.60 \times 10^{-19} \text{ C/electron}} \\ &= \boxed{6.2 \times 10^{18} \text{ electrons}}\end{aligned}$$

Insight

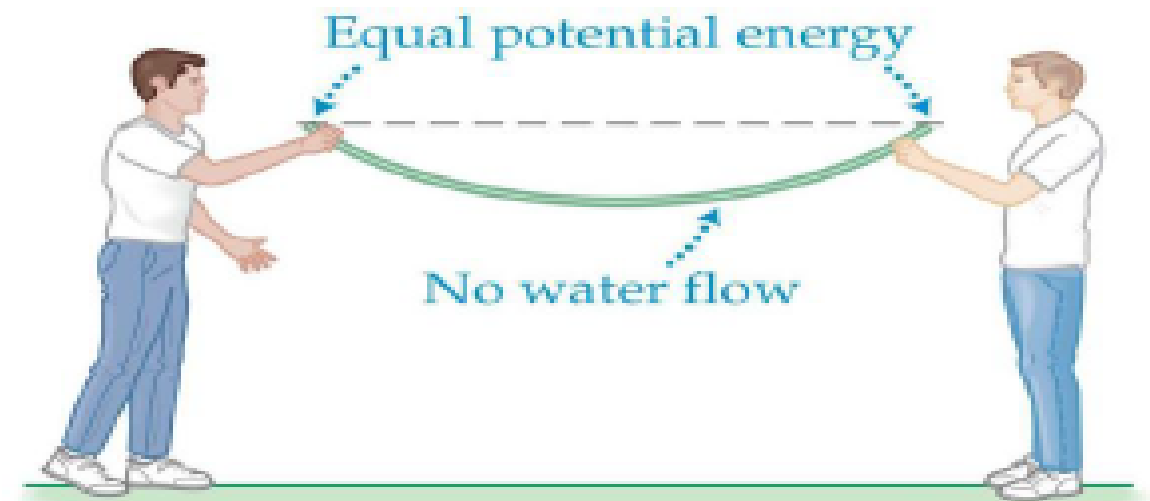
Thus, even a modest current flowing for a brief time corresponds to the transport of an extremely large number of electrons.

Electric Current, Resistance, and Semiconductors

- When charge flows through a closed path and returns to its starting point, we say that the closed path is an electric circuit.
- In a type of circuit known as a direct-current circuit, or DC circuit, the current always flows in the same direction. Circuits that run on batteries are typically DC circuits.
- Circuits with currents that periodically reverse their direction are referred to as alternating-current circuits, or AC circuits. The electricity provided by a wall plug in your house is AC.

Electric Current, Resistance, and Semiconductors

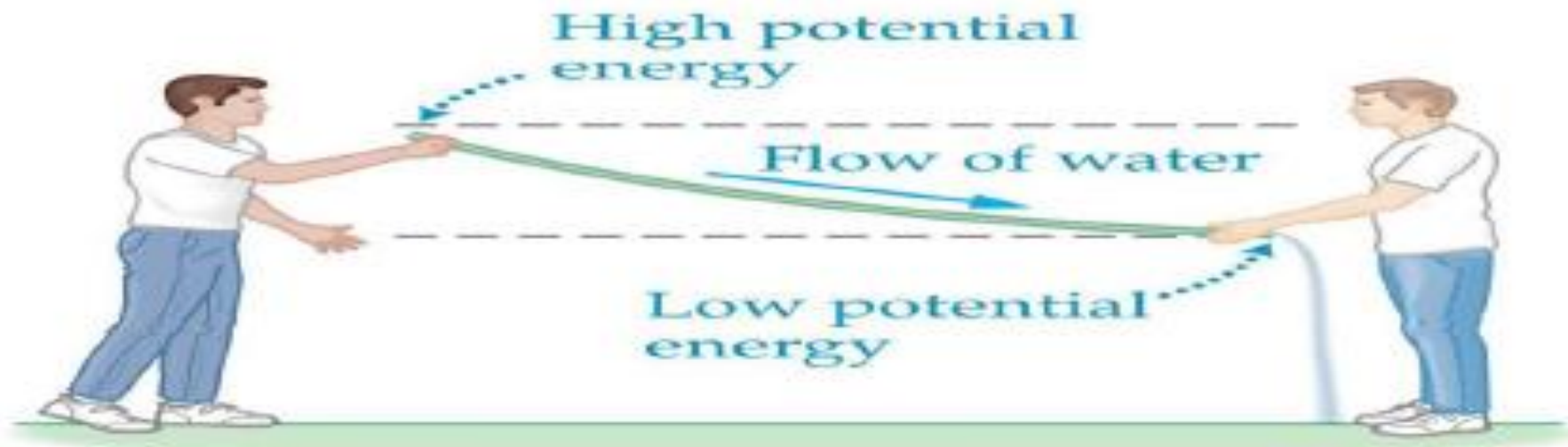
- Although electrons move fairly freely in metal wires, something has to push on them to get them going and keep them going. It's like water in a garden hose; the water flows only when a force pushes on it. Similarly, electrons flow in a circuit only when an electrical force pushes on them.
- Figure (a) below shows that there is no water flow if both ends of the garden hose are held at the same level.



(a) Equal potential energy → no flow

Electric Current, Resistance, and Semiconductors

- Figure (b) shows that water flows from the end where the gravitational potential energy is high to the end where it is low. The difference in gravitational potential energy between the two ends of the hose results in a force on the water—which in turn produces a flow. A battery performs a similar function in an electric circuit.



(b) Water flows from high potential energy to low.

Electric Current, Resistance, and Semiconductors

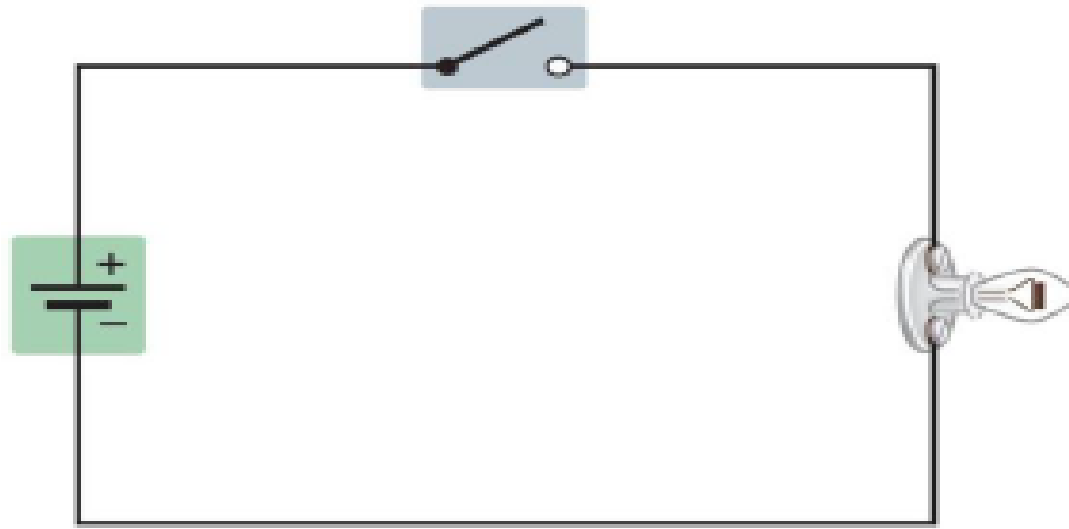
- A battery uses **chemical reactions** to produce a difference in electric potential between its two ends, which are referred to as the terminals. The symbol for a battery is 
- A battery's **positive terminal** has a **high electrical potential** and is denoted with a plus (+) sign; the **negative terminal** has a **low electric potential** and is denoted with a minus sign (-).

Electric Current, Resistance, and Semiconductors

- When a battery is connected to a circuit, electrons move in a closed path from one terminal of the battery through the circuit and back to the other terminal of the battery. The electrons leave from the negative terminal of the battery and return to the positive terminal.
- The situation is similar to the flow of blood in your body. Your heart acts like a battery, causing blood to flow through a closed circuit of arteries and veins in your body.

Electric Current, Resistance, and Semiconductors

- The circuit diagram in figure (b) below shows that the switch is open—creating an open circuit. When a circuit is open, no charge can flow. When the switch is closed, electrons flow through the circuit and the light glows.





Electric Current, Resistance, and Semiconductors

- The **difference in electric potential** between the terminals of the battery is the **electromotive force**, or emf. Symbolically, the electromotive force is represented by the symbol ε (the Greek letter epsilon). The unit of emf is the same as that of electrical potential, namely, the volt.
- The emf determines the amount of work a battery does to move a certain amount of charge around a circuit.

Electric Current, Resistance, and Semiconductors

- To be specific, the magnitude of the work done by a battery with the emf ε as charge ΔQ moves from one terminal to the other is

Work Done by a Battery

work done by a battery = amount of charge $\times$ emf of battery

$$W = (\Delta Q)\varepsilon$$

SI unit: J

Electric Current, Resistance, and Semiconductors

- The following example illustrates how the charge that passes through a circuit and the work done by the battery moving that charge can be determined.

ACTIVE Example 21.3 Determine the Charge and the Work

A battery with an emf of 1.5 V delivers a current of 0.44 A to a flashlight bulb for 64 s (see Figure 21.3). Find **(a)** the charge that passes through the circuit and **(b)** the work done by the battery.

Solution (Perform the calculations indicated in each step.)

Part (a)

1. Rearrange the definition of current, $I = \Delta Q / \Delta t$, to solve for the charge that flows through the circuit: $\Delta Q = 28 \text{ C}$

Part (b)

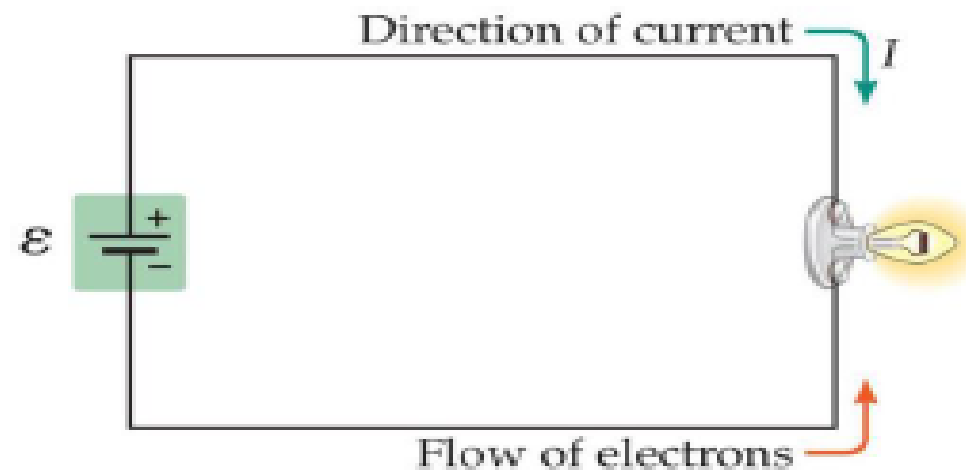
2. Substitute the value for ΔQ determined in part (a) in $W = \Delta Q \varepsilon$ to find the work done by the battery: $W = 42 \text{ J}$

Insight

Notice that the more charge a battery moves through a circuit, the more work it does. Similarly, the greater the emf, the greater the work. We can see, then, that a car battery that operates at 12 V and delivers several amps of current does much more work than a flashlight battery operating at 1.5 V and delivering 0.44 A—as expected.

Electric Current, Resistance, and Semiconductors

- When drawing an electric circuit, it's helpful to include an arrow to indicate the flow of current. By convention, the direction of the current in an electric circuit is the direction in which a positive test charge would move.
- In typical circuits, the charges that flow are actually negatively charged electrons. As a result, the flow of electrons and the current arrow point in opposite directions, as indicated in the figure below.



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Electric Current, Resistance, and Semiconductors

- To move electrons against the resistance of a wire, it is necessary to apply a potential difference between the wire's ends.
- Ohm's law relates the applied potential difference to the current produced and the wire's resistance. To be specific, the three quantities are related as follows:

Ohm's Law

applied voltage = current $\times$ resistance

$$V = IR$$

SI unit: volt (V)

Electric Current, Resistance, and Semiconductors

- Ohm's law is named for the German physicist Georg Simon Ohm (1789–1854).
- Rearranging Ohm's law to solve for the resistance, we find

$$R = V/I$$

- From this expression, it is clear that resistance has units of volts per amp. A resistance of 1 volt per amp defines a new unit—the *ohm*. The Greek letter omega (Ω) is used to designate the ohm. Thus,

$$1 \Omega = 1 \text{ V/A}$$

- A device for measuring resistance is called an ohmmeter.

Electric Current, Resistance, and Semiconductors

- A resistor is a small device used in electric circuits to provide a particular resistance to current. The resistance of a resistor is given in ohms, as shown in the following Quick Example.

QUICK Example 21.4 What's the Current?

A potential difference of 24 V is applied to a 150-Ω resistor. How much current flows through the resistor?

Solution

Solving Ohm's law for the current, we find

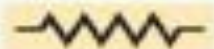
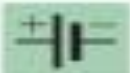
$$I = \frac{V}{R} = \frac{24 \text{ V}}{150 \text{ } \Omega} = \frac{24 \text{ V}}{150 \text{ V/A}} = \boxed{0.16 \text{ A}}$$

In an electric circuit, a resistor is signified by  a zigzag line.

Electric Current, Resistance, and Semiconductors

- The following chart summarizes the elements of electric circuits, their symbols, and their physical characteristics.

Table 21.1 Elements of Electric Circuits

Circuit Element	Symbol	Physical Characteristics
Resistor		Resists the flow of electric current. Converts electric energy into thermal energy.
Ideal wire		An ideal wire has zero resistance. It is used to connect various elements in a circuit.
Battery		A device that produces a constant difference in electrical potential between its two terminals.
Switches (open and closed)		Devices used to control whether electric current is allowed to flow through a circuit or a portion of a circuit.
Diode		A device that allows electric current to flow in one direction only.
Incandescent lightbulb		A device containing a resistor that gets hot enough to give off visible light.

Electric Current, Resistance, and Semiconductors

- A wire's resistance is affected by several factors.
- The resistance of a wire depends on the **material** from which it is made. For example, if a wire is made of copper, its resistance is less than if it is made from iron. The resistance of a given material is described by its resistivity, ρ .
- A wire's resistance also depends on its **length**, L , and its **cross-sectional area**, A . To understand these factors, let's consider water flowing through a hose. If the hose is very long, its resistance to the water is correspondingly large. On the other hand, a wide hose, with a greater cross-sectional area, offers less resistance to the water.

Electric Current, Resistance, and Semiconductors

- Combining these observations regarding the factors that affect a wire's resistance, we can write the following relationship:

Definition of Resistivity, ρ

resistance = resistivity $\times$ $\frac{\text{length of wire}}{\text{cross-sectional area}}$

$$R = \rho \left(\frac{L}{A} \right)$$

SI units: ohm-meters ($\Omega \cdot \text{m}$)

- The units of resistivity are ohm-meters ($\Omega \cdot \text{m}$), and its magnitude varies greatly with the type of material. Insulators have large/high resistivities; conductors have low resistivities.

Electric Current, Resistance, and Semiconductors

- As a wire is heated, its resistivity tends to increase. This effect occurs because atoms that are jiggling more rapidly are more likely to collide with electrons and slow their progress through the wire.
- The following table summarizes the four factors that affect the resistance of a wire.

Table 21.2 Factors Affecting the Resistance of a Wire

Factor	Example of Lower Resistance	Example of Higher Resistance	Reason for Difference
Material	 Copper	 Lead	Resistance depends on the material. The lead wire has a greater resistance than the copper wire.
Length			Resistance increases with length. A long wire has a greater resistance than a short wire.
Area			Resistance increases as cross-sectional area decreases. A small-diameter wire has a greater resistance than a large-diameter wire.
Temperature			Resistance increases as temperature increases. A heated wire has a greater resistance than a cooled wire.

Electric Current, Resistance, and Semiconductors

- Though Ohm's law is an excellent approximation for metal wires and the resistors used in electric circuits, it does not apply to all materials. **Materials known as semiconductors are an important exception to Ohm's law.**
- Elements such as germanium and silicon are insulators in their pure form. However, when impurities are added—which is referred to as **doping**—these substances can conduct electricity. Doping produces two types of semiconductors.

Electric Current, Resistance, and Semiconductors

- If a small amount of arsenic is added to silicon—say, one arsenic atom per million silicon atoms—the silicon becomes a conductor. The arsenic-doped silicon conducts electricity because electrons break free from the arsenic atoms and move freely through the material.
- Silicon doped in this way is referred to as an **n-type semiconductor** because current is carried by negative (n) electrons.

Electric Current, Resistance, and Semiconductors

- Semiconductors can be used to make a variety of electronic devices. The simplest semiconducting device, the diode, consists of a p-type semiconductor joined to an n-type semiconductor. A diode is shown in the figure below.



Physical appearance



Semiconductor types

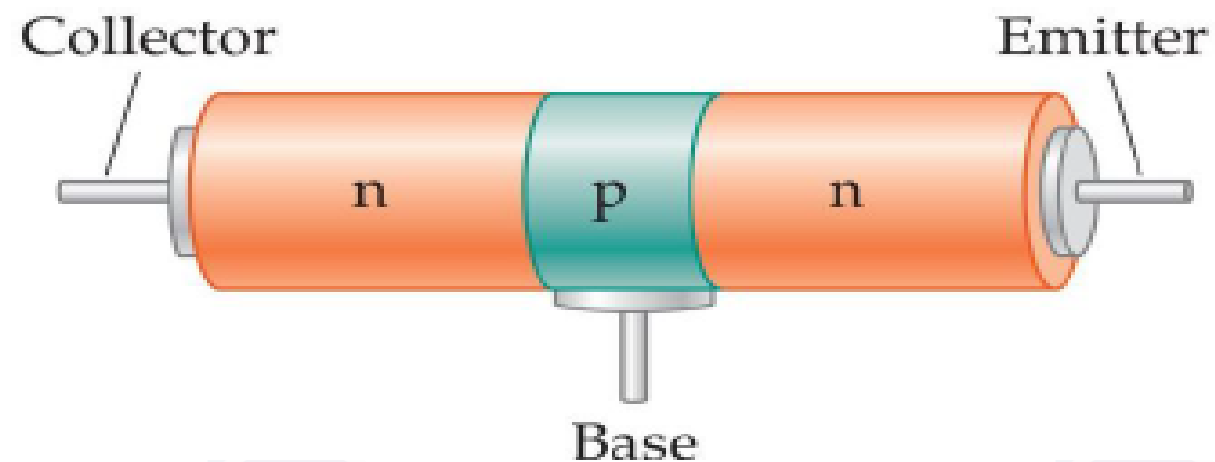


Circuit symbol

- The basic property of a diode is that it allows current to flow in one direction, but not the other.

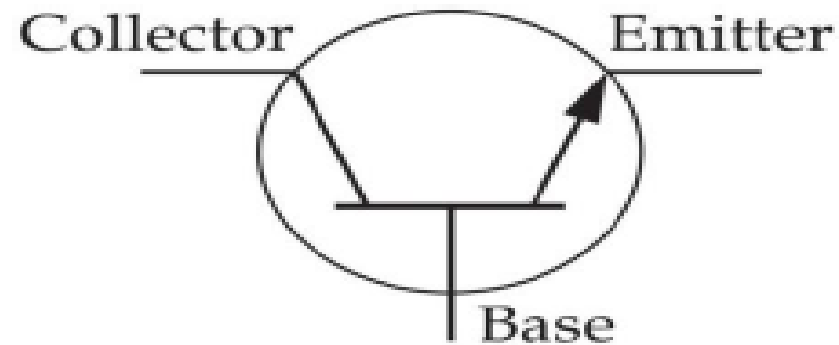
Electric Current, Resistance, and Semiconductors

- Another useful semiconductor device is produced by making a "sandwich" of three layers of semiconductors. The most common type of transistor has an n-type semiconductor on either side of the sandwich and a thin p-type semiconductor in the middle, as is shown in the figure below. This is known as an npn transistor.



Electric Current, Resistance, and Semiconductors

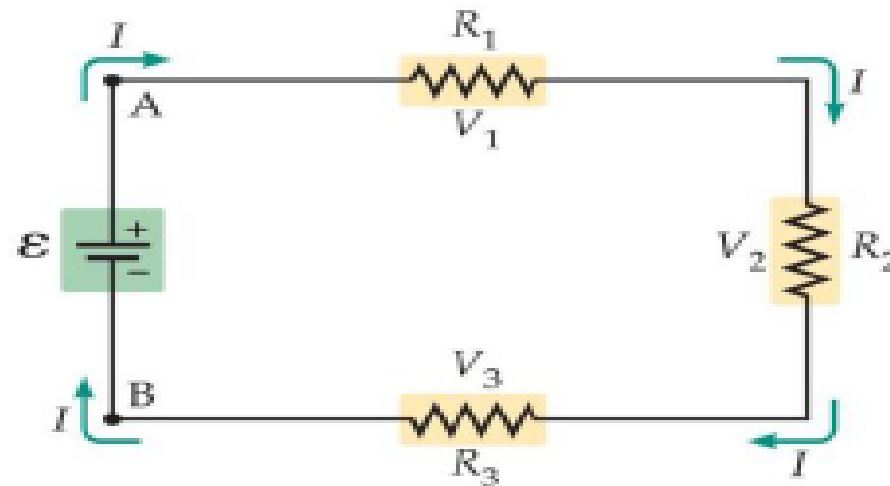
- Transistors can also be made with the opposite sequence of semiconductors, resulting in a pnp transistor.
- The basic function of a transistor is to act as an electronic switch that controls the flow of current in a circuit.



(b)

Electric Circuits

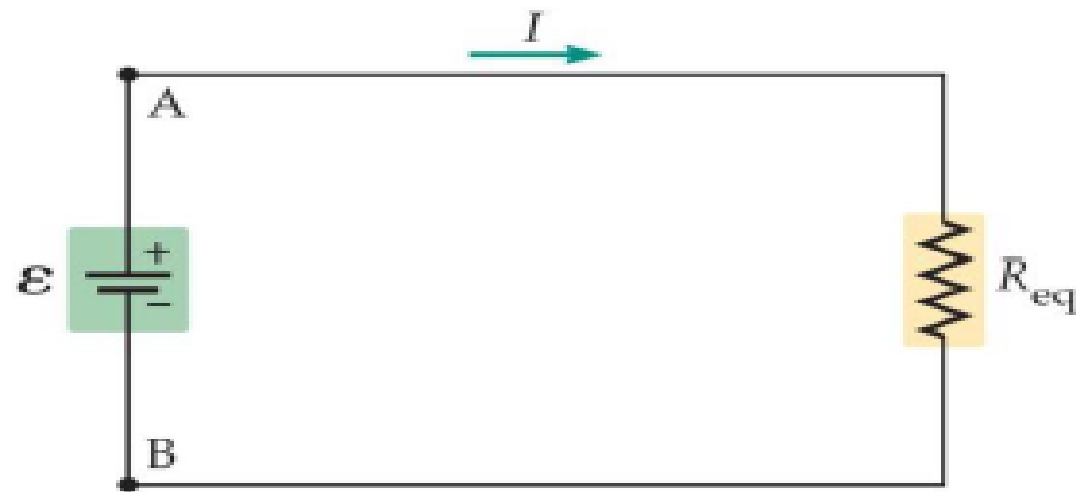
- Electric circuits often contain a number of resistors connected in various ways.
- One way resistors can be connected is end to end. Resistors connected in this way are said to form a series circuit. The figure below shows three resistors R_1 , R_2 , and R_3 , connected in series.



(a) Three resistors in series

Electric Circuits

- The three resistors acting together have the same effect—that is, they draw the same current—as a single resistor, which is referred to as the equivalent resistor, R_{eq} .
- This equivalence is illustrated in the figure below.



(b) Equivalent resistance has the same current.

- The equivalent resistor has the same current, I , flowing through it as each resistor in the original circuit.

Electric Circuits

- When resistors are connected in series, the equivalent resistance is simply the sum of the individual resistances.
- In our case, with three resistors, we have
$$R_{eq} = R_1 + R_2 + R_3$$
- In general, the equivalent resistance of resistors in series is the sum of all the resistances that are connected together:

Equivalent Resistance for Resistors in Series

equivalent resistance = resistance 1 + resistance 2 + resistance 3 + ...

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

SI unit: ohm (Ω)



Electric Circuits

- The equivalent resistance is greater than the greatest resistance of any individual resistor.
- In general, the more resistors connected in series, the greater the equivalent resistance.
- For example, the equivalent resistance of a circuit with two identical resistors, R , connected in series is $R_{eq} = R + R = 2R$. Thus, connecting two identical resistors in series produces an equivalent resistance that is twice the individual resistances.

Electric Circuits

- The following example illustrates the functioning of a series circuit.

GUIDED Example 21.5 | Three Resistors in Series

Series Circuit

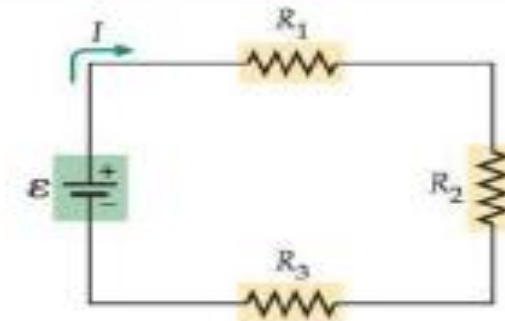
A circuit consists of three resistors connected in series to a 24.0-V battery. The current in the circuit is 0.0320 A. Given that $R_1 = 250.0 \, \Omega$ and $R_2 = 150.0 \, \Omega$, find (a) the resistance of R_3 and (b) the potential difference across each resistor.

Picture the Problem

The circuit is shown in our sketch. Notice that the same current, $I = 0.0320 \, \text{A}$, flows through each of the three resistors. This is the key characteristic of a series circuit.

Strategy

(a) First, we can find the equivalent resistance of the circuit by using Ohm's law, $R_{\text{eq}} = \mathcal{E}/I$. Since the resistors are in series, we also know that $R_{\text{eq}} = R_1 + R_2 + R_3$. We can solve this relation for the only unknown, R_3 . (b) We can then calculate the potential difference across each resistor using Ohm's law, $V = IR$.



Known

$\mathcal{E} = 24.0 \, \text{V}$
 $I = 0.0320 \, \text{A}$
 $R_1 = 250.0 \, \Omega$
 $R_2 = 150.0 \, \Omega$

Unknown

(a) $R_3 = ?$
 (b) $V_1 = ?$
 $V_2 = ?$
 $V_3 = ?$

Electric Circuits

GUIDED Example 21.5 | Three Resistors in Series (Continued)

Series Circuit

Solution

Part (a)

- 1 Use Ohm's law to find the equivalent resistance of the circuit:

$$R_{\text{eq}} = \frac{\mathcal{E}}{I} = \frac{24.0 \text{ V}}{0.0320 \text{ A}} \\ = 750 \, \Omega$$

Math HELP
Solving Simple
Equations
See Math
Review,
Section III

- 2 Set R_{eq} equal to the sum of the individual resistances, and solve for R_3 :

$$R_{\text{eq}} = R_1 + R_2 + R_3 \\ R_3 = R_{\text{eq}} - R_1 - R_2 \\ = 750 \, \Omega - 250.0 \, \Omega - 150.0 \, \Omega \\ = \boxed{350 \, \Omega}$$

Part (b)

- 3 Use Ohm's law to calculate the potential difference across R_1 :

$$V_1 = IR_1 \\ = (0.0320 \text{ A})(250.0 \, \Omega) \\ = \boxed{8.00 \text{ V}}$$

- 4 Find the potential difference across R_2 :

$$V_2 = IR_2 \\ = (0.0320 \text{ A})(150.0 \, \Omega) \\ = \boxed{4.80 \text{ V}}$$

- 5 Find the potential difference across R_3 :

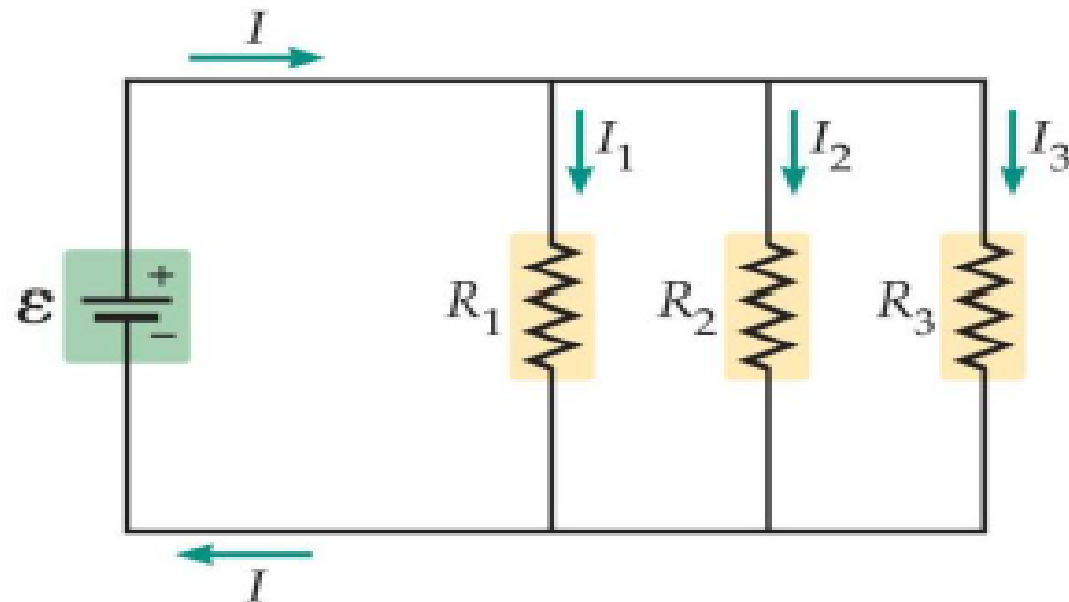
$$V_3 = IR_3 \\ = (0.0320 \text{ A})(350 \, \Omega) \\ = \boxed{11.2 \text{ V}}$$

Insight

Notice that the greater the resistance, the greater the potential difference. In addition, the sum of the individual potential differences is the voltage of the battery, $8.00 \text{ V} + 4.80 \text{ V} + 11.2 \text{ V} = 24.0 \text{ V}$, as expected.

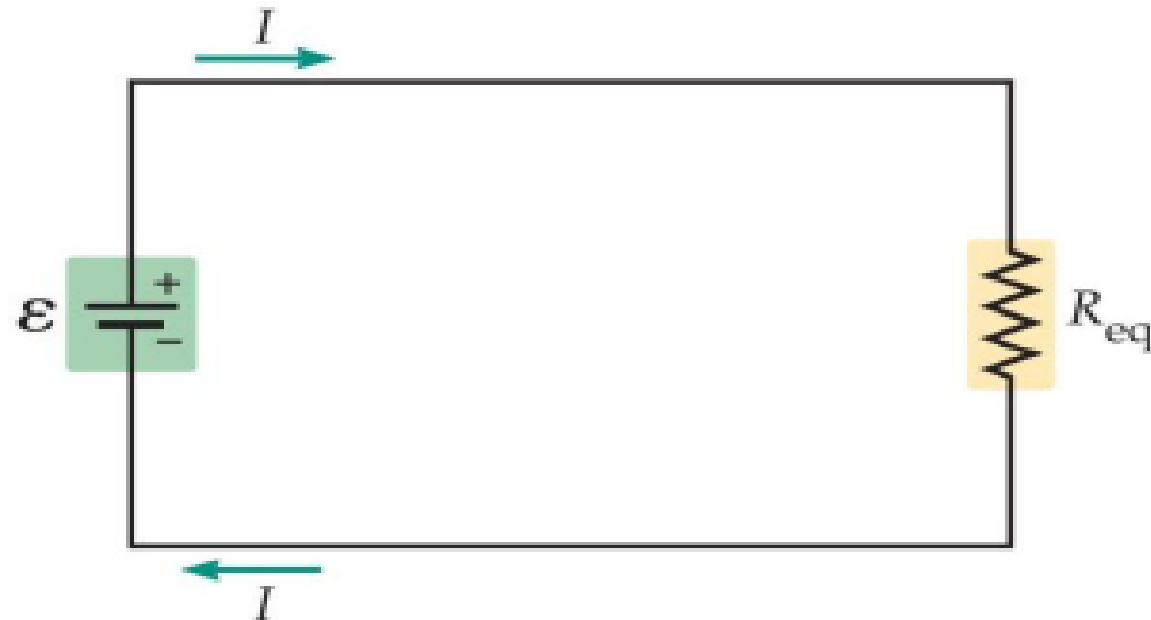
Electric Circuits

- Resistors that are connected across the same potential difference are said to form a parallel circuit.
- An example of three resistors connected in parallel is shown the figure below.



Electric Circuits

- In a case like this, the electrons have three parallel paths through which they can flow—like parallel lanes on the highway.
- The three resistors acting together draw the same current as a single equivalent resistor, R_{eq} , as indicated in the figure below.



Electric Circuits

- When resistors are connected in parallel, the reciprocal of the equivalent resistance is equal to the sum of the reciprocals of the individual resistances. Thus, for our circuit of three resistors, we have

$$1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$$

- In general, the inverse equivalent resistance is equal to the sum of all of the individual inverse resistances:

Electric Circuits

Equivalent Resistance for Resistors in Parallel

$$\frac{1}{\text{equivalent resistance}} = \frac{1}{\text{resistance 1}} + \frac{1}{\text{resistance 2}} + \frac{1}{\text{resistance 3}} + \dots$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

SI unit: ohm (Ω)

- As an example of parallel resistors, consider a circuit with two identical resistors, R , connected in parallel. The equivalent resistance in this case is

$$1/R_{\text{eq}} = 1/R + 1/R$$

$$1/R_{\text{eq}} = 2/R$$



Electric Circuits

- Solving for the equivalent resistance gives $R_{eq} = \frac{1}{2}R$. Thus, connecting two identical resistors in parallel produces an equivalent resistance that is half of the individual resistances.
- A similar calculation shows that three resistors, R , connected in parallel produces an equivalent that is one-third of the original resistances, or $R_{eq} = \frac{1}{3}R$.
- These results show a clear trend, namely, the more resistors connected in parallel, the smaller the equivalent resistance.

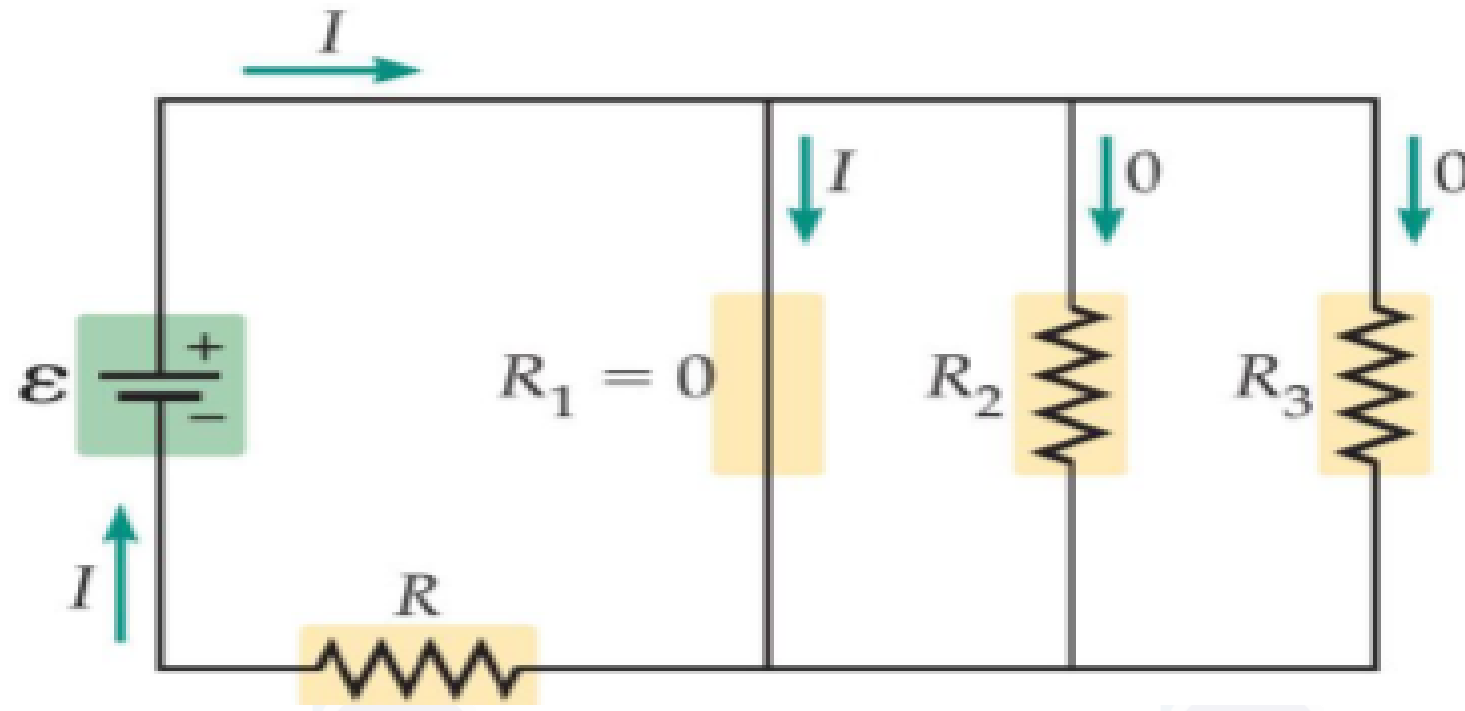


Electric Circuits

- In general, the equivalent resistance of a parallel circuit is less than or equal to the smallest individual resistance. What happens if one of the individual resistances is zero?
- In this case, the equivalent resistance is also zero, because R_{eq} is less than or equal to the smallest individual resistance, and a resistance can't be negative.

Electric Circuits

- This situation, referred to as a short circuit, is illustrated in the figure below. In a short circuit, all the current flows through the path of zero resistance.



Electric Circuits

- The following example illustrates the functioning of a parallel circuit.

GUIDED Example 21.6 | Three Resistors in Parallel

Parallel Circuit

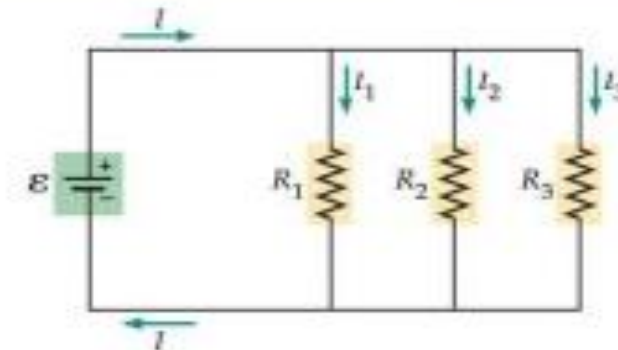
Consider a circuit with three resistors, $R_1 = 250.0 \, \Omega$, $R_2 = 150.0 \, \Omega$, and $R_3 = 350.0 \, \Omega$, connected in parallel with a 24.0-V battery. Find (a) the total current supplied by the battery and (b) the current through each resistor.

Picture the Problem

Our sketch indicates the parallel connection of the resistors with the battery. Notice that each of the resistors experiences precisely the same potential difference—namely, the 24.0 V produced by the battery. This is the feature that characterizes parallel connections.

Strategy

(a) We can find the total current by solving Ohm's law, $V = IR$, for the current. This gives $I = \varepsilon/R_{\text{eq}}$, where $\varepsilon = V = 24.0 \, \text{V}$ and $1/R_{\text{eq}} = 1/R_1 + 1/R_2 + 1/R_3$. (b) For each resistor the current is given by Ohm's law, $I = \varepsilon/R$.



Known

$\varepsilon = 24.0 \, \text{V}$
 $R_1 = 250.0 \, \Omega$
 $R_2 = 150.0 \, \Omega$
 $R_3 = 350.0 \, \Omega$

Unknown

(a) $I = ?$
 (b) $I_1 = ?$
 $I_2 = ?$
 $I_3 = ?$

Electric Circuits

GUIDED Example 21.6 | Three Resistors in Parallel (Continued)

Parallel Circuit

Solution

Part (a)

1 Find the equivalent resistance of the circuit:

$$\begin{aligned}\frac{1}{R_{\text{eq}}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ &= \frac{1}{250.0 \, \Omega} + \frac{1}{150.0 \, \Omega} + \frac{1}{350.0 \, \Omega} \\ &= 0.01352 \, \Omega^{-1} \\ R_{\text{eq}} &= \frac{1}{0.01352 \, \Omega^{-1}} = 73.96 \, \Omega\end{aligned}$$

2 Use Ohm's law to find the total current:

$$I = \frac{\mathcal{E}}{R_{\text{eq}}} = \frac{24.0 \, \text{V}}{73.96 \, \Omega} = \boxed{0.325 \, \text{A}}$$

Part (b)

3 Calculate I_1 using $I_1 = \mathcal{E}/R_1$ with $\mathcal{E} = 24.0 \, \text{V}$:

$$I_1 = \frac{\mathcal{E}}{R_1} = \frac{24.0 \, \text{V}}{250.0 \, \Omega} = \boxed{0.0960 \, \text{A}}$$

4 Repeat the calculation of Step 3 for resistors 2 and 3:

$$I_2 = \frac{\mathcal{E}}{R_2} = \frac{24.0 \, \text{V}}{150.0 \, \Omega} = \boxed{0.160 \, \text{A}}$$

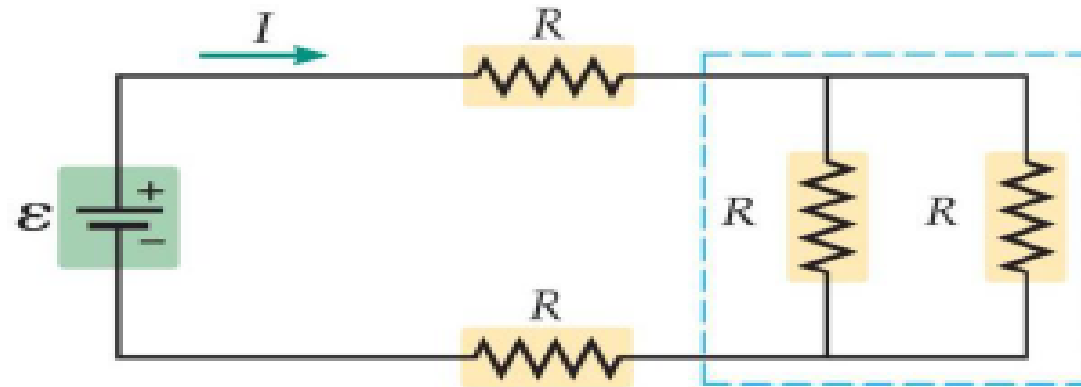
$$I_3 = \frac{\mathcal{E}}{R_3} = \frac{24.0 \, \text{V}}{350.0 \, \Omega} = \boxed{0.0686 \, \text{A}}$$

Insight

As expected, the smallest resistor, R_2 , carries the greatest current. The three currents combined yield the total current, as they must. That is, $I_1 + I_2 + I_3 = 0.0960 \, \text{A} + 0.160 \, \text{A} + 0.0686 \, \text{A} = 0.325 \, \text{A} = I$. Also, notice that the equivalent resistance ($73.96 \, \Omega$) is less than that of the smallest resistor in the parallel circuit ($150.0 \, \Omega$).

Electric Circuits

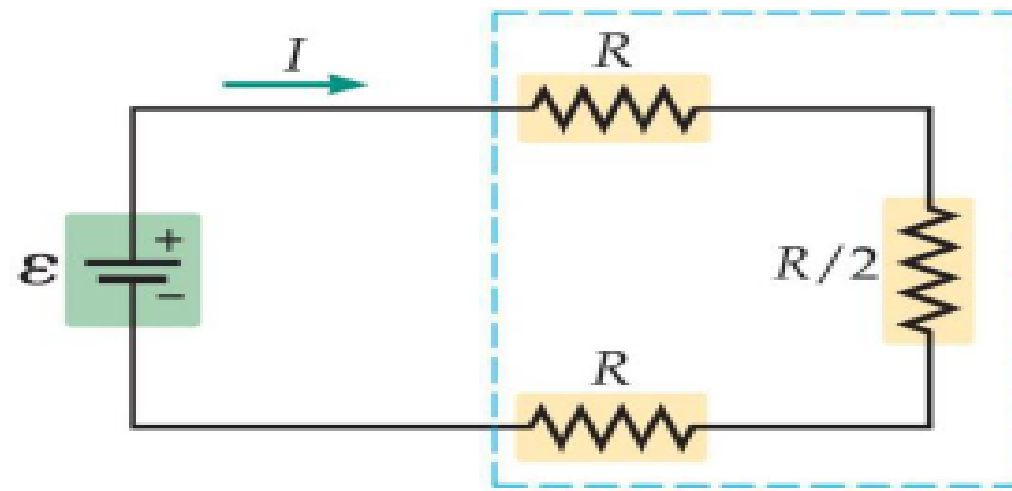
- The rules that apply for series and parallel resistors can be applied to a variety of interesting circuits that aren't purely series or parallel.
- The circuit in the figure below contains a total of four resistors, each with resistance R , connected in a way that combines series and parallel features. Because the circuit is not strictly series or parallel, we can't directly calculate the equivalent resistance.



(a) Replace parallel resistors

Electric Circuits

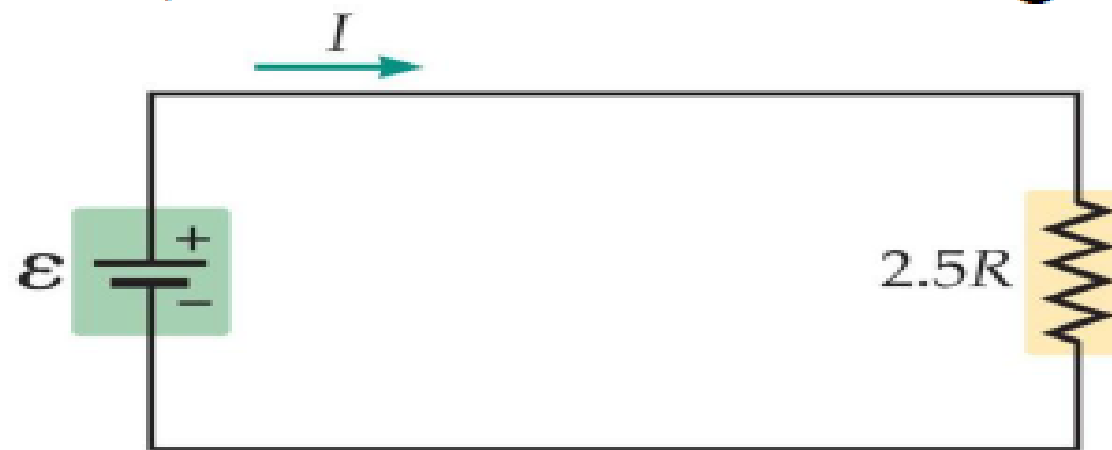
- What we can do, however, is break the circuit into smaller subcircuits, each of which is purely series or purely parallel. For example, we first note that the two vertically oriented resistors on the right are in parallel with one another; hence they can be replaced with their equivalent resistance $R/2$.
- The next step is to replace these two resistors with $R/2$. This yields the circuit shown below.



(b) Replace series resistors

Electric Circuits

- Notice that this equivalent circuit consists of three resistors in series, R , $\frac{1}{2}R$, and R . The equivalent resistance of these resistors is equal to their sum, $R_{eq} = R_1 + R_2 + R_3 = 2.5R$.
- Therefore, the equivalent resistance of the original circuit is $2.5R$, as indicated in the figure below.



(c) Final equivalent resistance

Electric Circuits

- By considering the resistors in pairs or groups that are connected in parallel or in series, you can reduce the entire circuit to one equivalent circuit. This method is applied in the following example.

GUIDED Example 21.7 | Combination Special

Combination Circuit

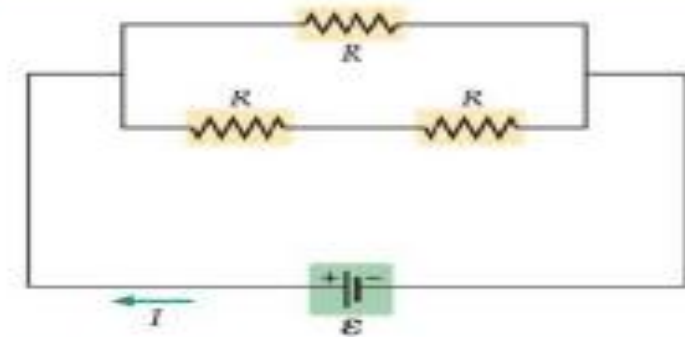
In the circuit shown in the diagram below, the emf of the battery is 12 V, and each resistor has a resistance of $450\ \Omega$. Find the current supplied by the battery to this circuit.

Picture the Problem

The circuit for this problem has three resistors connected to a battery. The lower two resistors are in series with one another, and they are in parallel with the upper resistor. The battery has an emf of 12 V.

Strategy

The current supplied by the battery is given by Ohm's law, $I = \mathcal{E}/R_{\text{eq}}$, where R_{eq} is the equivalent resistance of the three resistors. To find R_{eq} we first note that the lower two resistors are in series, giving a net resistance of $2R$. Next, the upper resistor, R , is in parallel with the pair of resistors having resistance $2R$. Calculating the equivalent resistance of this combination yields the desired R_{eq} .



Known

$$\mathcal{E} = 12\text{ V}$$

$$R = 450\ \Omega$$

Unknown

$$I = ?$$

Electric Circuits

GUIDED Example 21.7 | Combination Special (Continued)

Combination Circuit

Solution

1 Calculate the equivalent resistance of the two lower resistors:

$$R_{\text{eq,lower}} = R + R = 2R$$

2 Use the result of Step 1 to calculate the equivalent resistance of R in parallel with $2R$:

$$\begin{aligned}\frac{1}{R_{\text{eq}}} &= \frac{1}{R} + \frac{1}{2R} = \frac{3}{2R} \\ R_{\text{eq}} &= \frac{2}{3}R \\ &= \frac{2}{3}(450 \, \Omega) \\ &= 300 \, \Omega\end{aligned}$$

Math HELP
Solving Simple Equations
See Math Review, Section III

3 Find the current supplied by the battery, I :

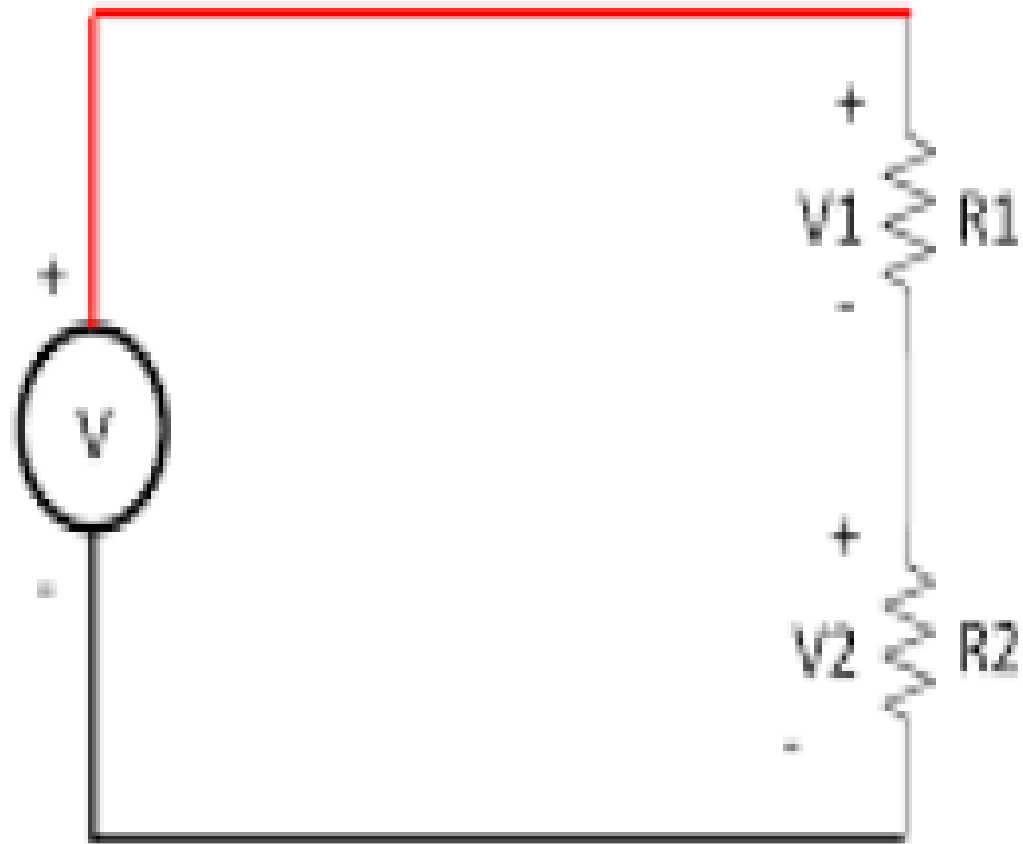
$$\begin{aligned}I &= \frac{\mathcal{E}}{R_{\text{eq}}} \\ &= \frac{12 \, \text{V}}{300 \, \Omega} \\ &= \boxed{0.040 \, \text{A}}\end{aligned}$$

Insight

Notice that the total resistance of the three $450\text{-}\Omega$ resistors is less than $450 \, \Omega$ —in fact, it is only $300 \, \Omega$.

Voltage divider rule

The voltage drop across any given resistor in a series circuit is equal to the ratio of that resistor to the total resistance, multiplied by source voltage.



V1 is the voltage across R1:

$$V1 = \frac{R1}{R1+R2} * V$$

V2 is the voltage across R2:

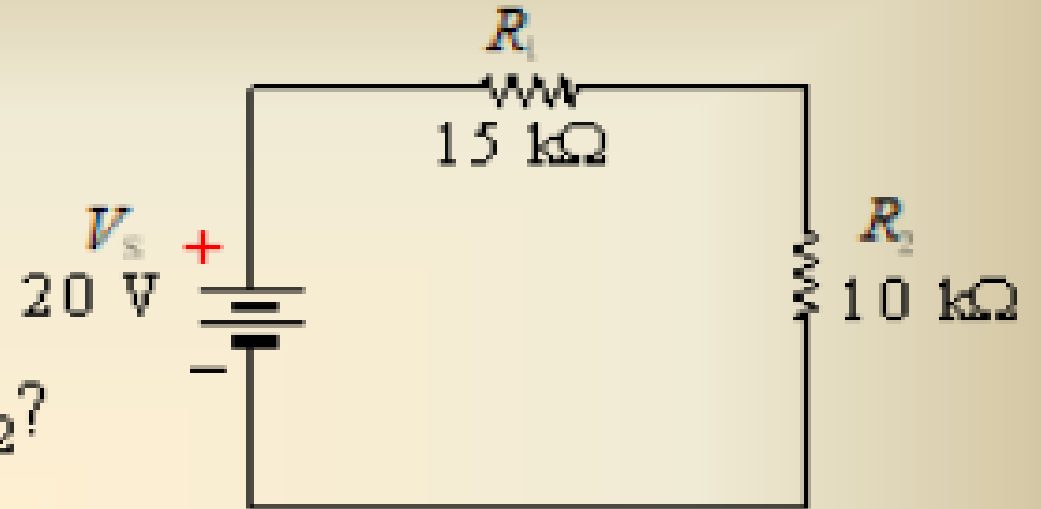
$$V2 = \frac{R2}{R1+R2} * V$$

Voltage division rule

Voltage divider

Example:

What is the voltage across R_2 ?



Solution:

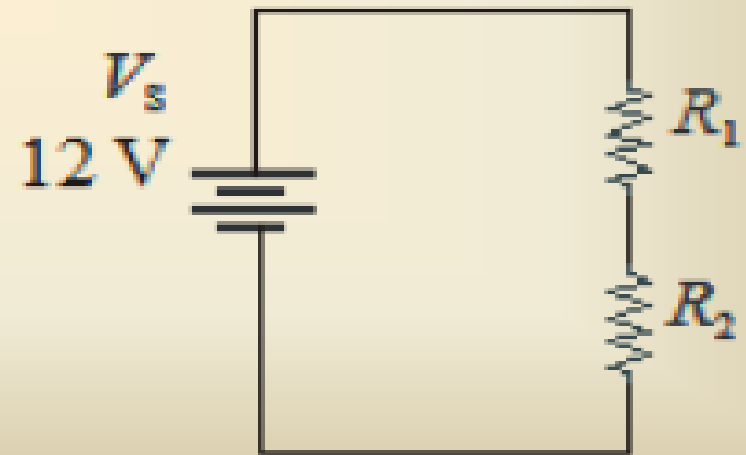
The total resistance is $25 \text{ k}\Omega$.

Applying the voltage divider formula:

$$V_2 = \left(\frac{R_2}{R_T} \right) V_s = \left(\frac{10 \text{ k}\Omega}{25 \text{ k}\Omega} \right) 20 \text{ V} = 8.0 \text{ V}$$

Voltage divider rule

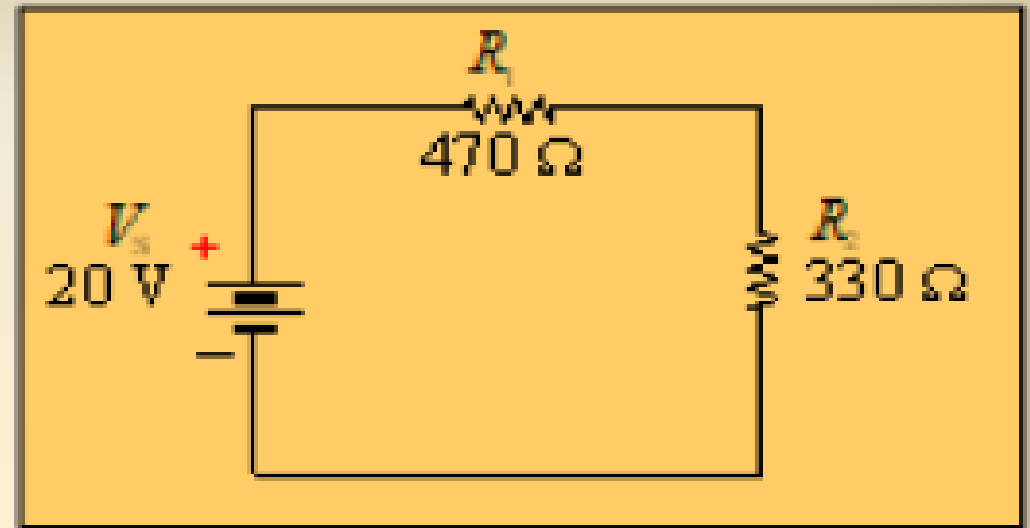
What is the voltage drop across resistors R_1 and R_2 in the diagram below if $R_1 = 4$ ohms and $R_2 = 10$ ohms?



Power in Series Circuits

Example:

Use the voltage divider rule to find V_1 and V_2 . Then find the power in R_1 and R_2 and P_T



Solution:

Applying the voltage divider rule:

$$V_1 = \left(\frac{470 \Omega}{800 \Omega} \right) 20 \text{ V} = 11.75 \text{ V}$$

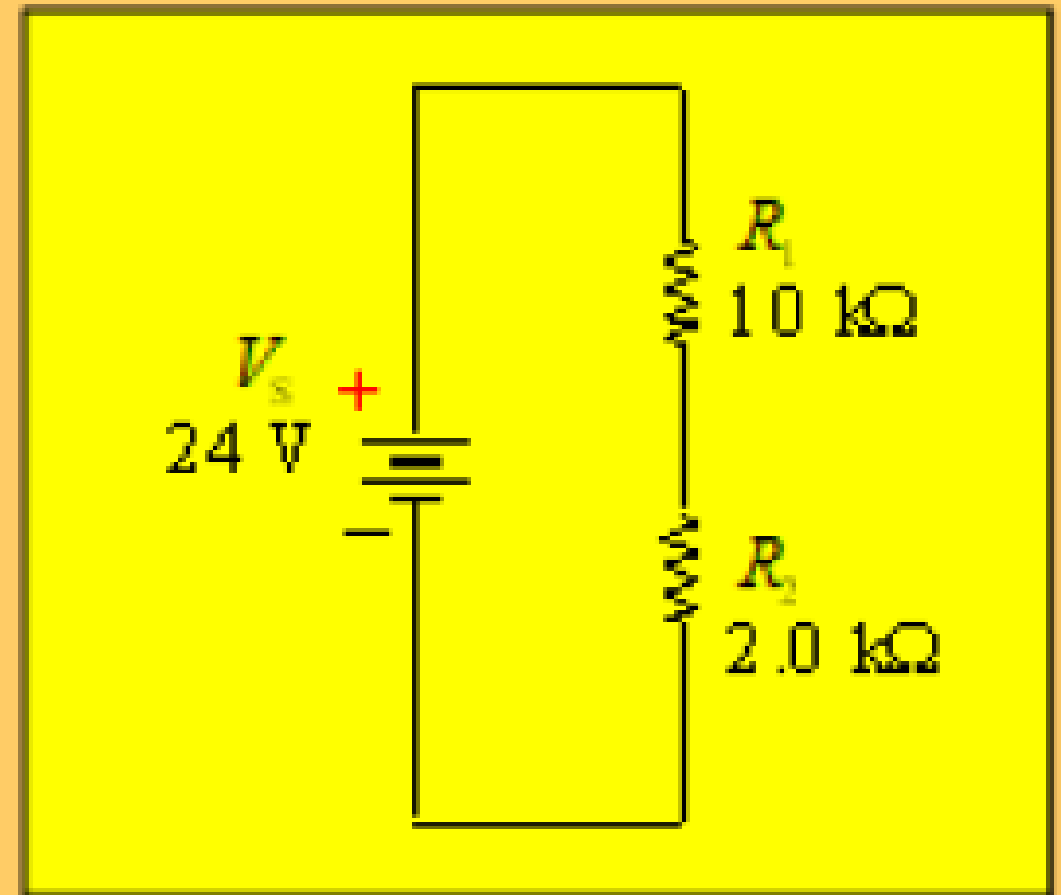
$$V_2 = \left(\frac{330 \Omega}{800 \Omega} \right) 20 \text{ V} = 8.25 \text{ V}$$

The power dissipated by each resistor is:

$$\left. \begin{aligned} P_1 &= \frac{(11.75 \text{ V})^2}{470 \Omega} = 0.29 \text{ W} \\ P_2 &= \frac{(8.25 \text{ V})^2}{330 \Omega} = 0.21 \text{ W} \end{aligned} \right\} P_T = 0.5 \text{ W}$$

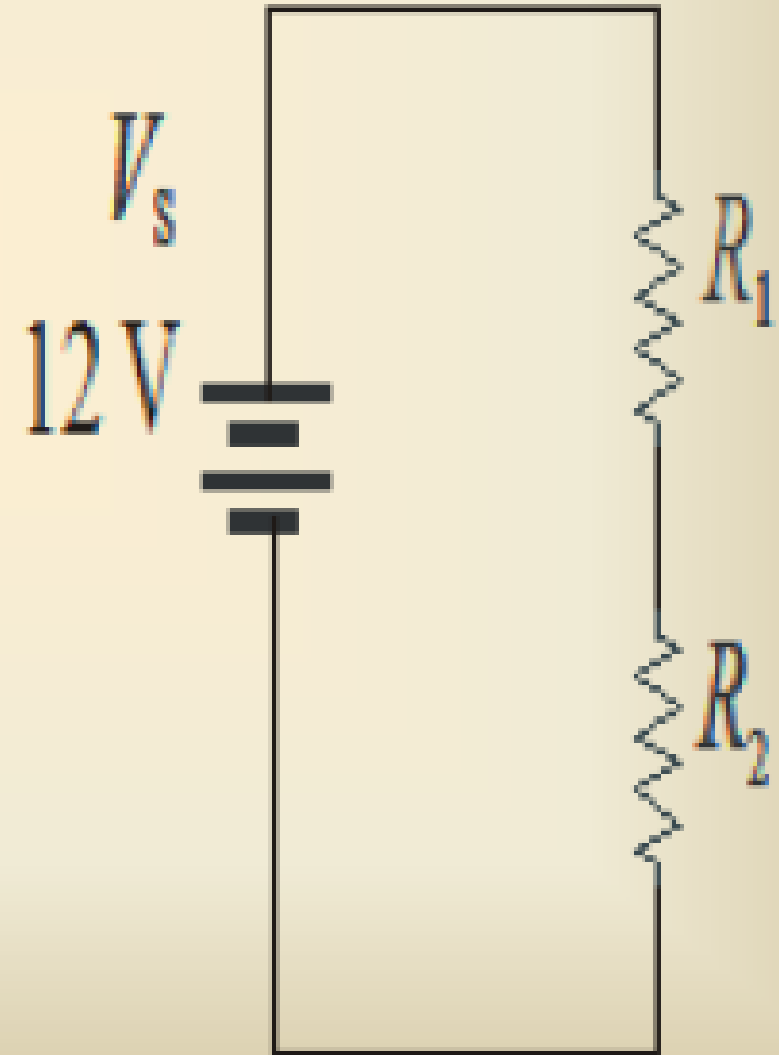
The current in the $10\text{ k}\Omega$ resistor is

- a. 0.5 mA
- b. 2.0 mA
- c. 2.4 mA
- d. 10 mA



Question:

Assume R_1 is twice the size of R_2 . What is the voltage across R_1 ?





Current divider

When current enters a node (junction) it divides into currents with values that are inversely proportional to the resistance values.

The most widely used formula for the current divider is the two-resistor equation. For resistors R_1 and R_2 ,

$$I_1 = \left(\frac{R_2}{R_1 + R_2} \right) I_T \quad \text{and} \quad I_2 = \left(\frac{R_1}{R_1 + R_2} \right) I_T$$

Notice the subscripts. The resistor in the numerator is not the same as the one for which current is found.

Example

Assume that R_1 is a $2.2 \text{ k}\Omega$ resistor that is in parallel with R_2 , which is $4.7 \text{ k}\Omega$. If the total current into the resistors is 8.0 mA , what is the current in each resistor?

Solution

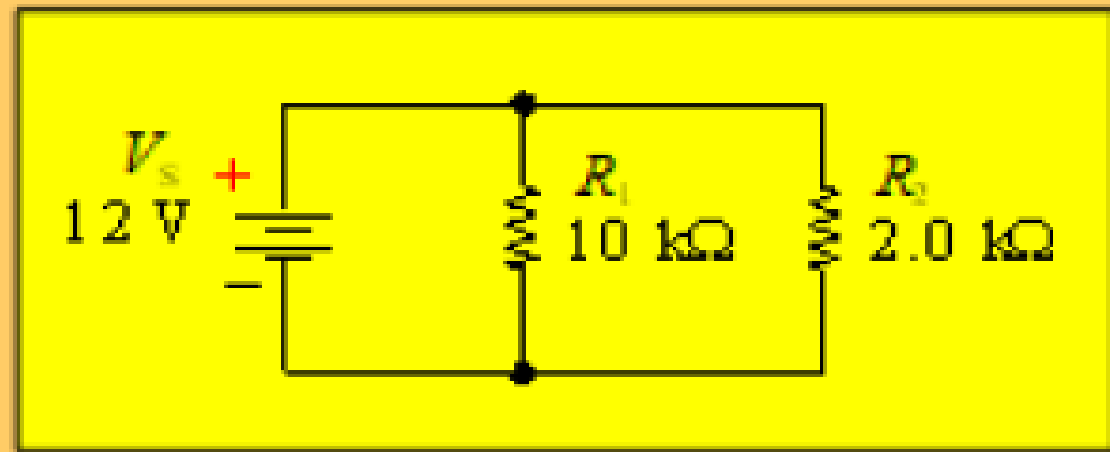
$$I_1 = \left(\frac{R_2}{R_1 + R_2} \right) I_T = \left(\frac{4.7 \text{ k}\Omega}{69 \text{ k}\Omega} \right) 8.0 \text{ mA} = 5.45 \text{ mA}$$

$$I_2 = \left(\frac{R_1}{R_1 + R_2} \right) I_T = \left(\frac{2.2 \text{ k}\Omega}{6.9 \text{ k}\Omega} \right) 8.0 \text{ mA} = 2.55 \text{ mA}$$

Notice that the *larger* resistor has the *smaller* current.

Quiz

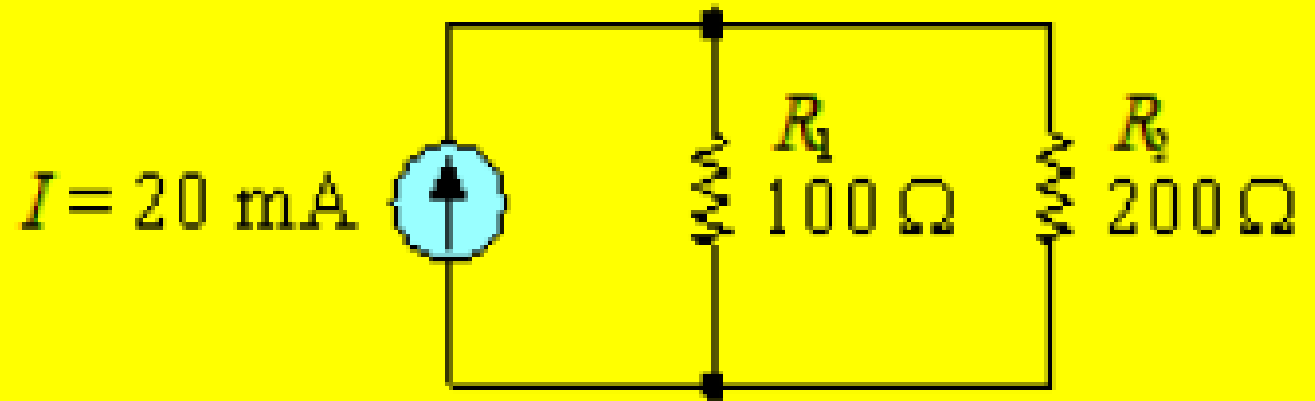
Find the total current leaving the source and the current in each resistor?



Quiz

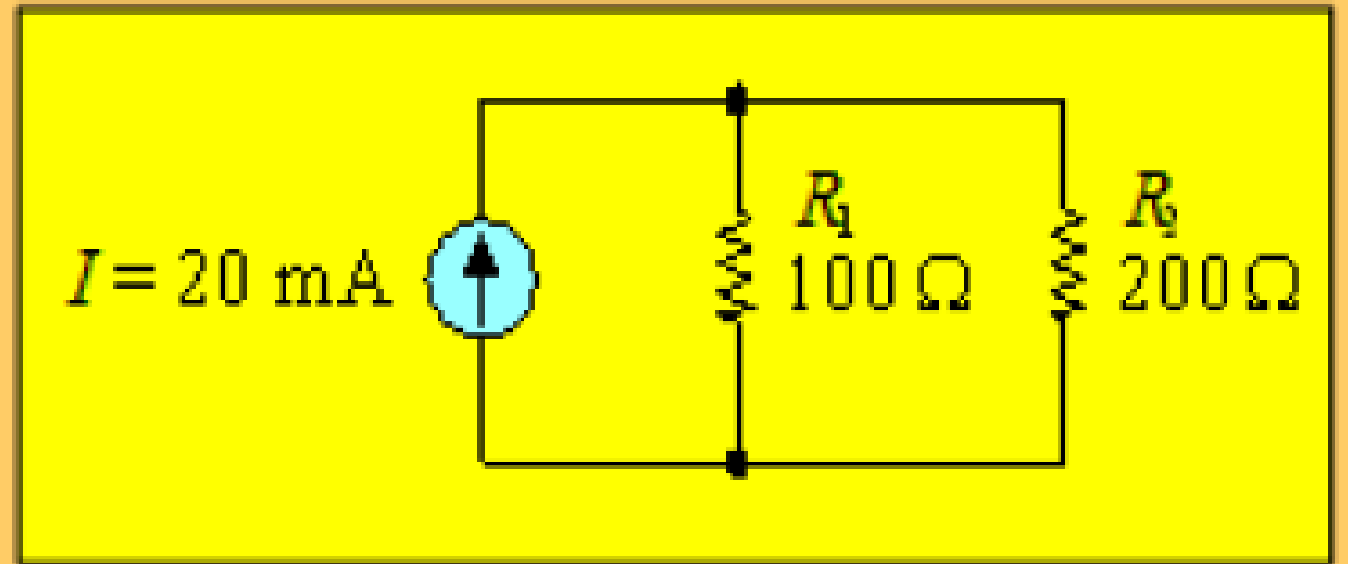
What is the value of the current in R_1 ?

- a. 6.7 mA
- b. 13.3 mA
- c. 20 mA
- d. 26.7 mA



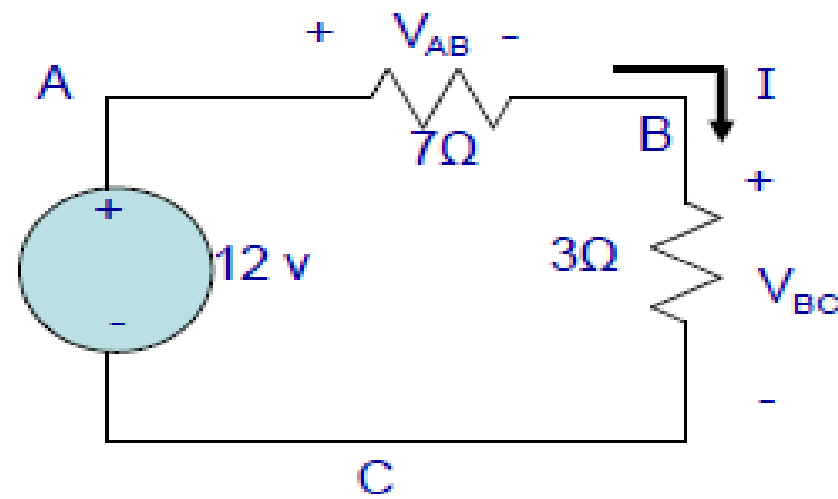
Find the voltage across R_2 ?

- a. 0 V
- b. 0.67 V
- c. 1.33 V
- d. 4.0 V



Circuit Analysis

- Since $V_{AB} = I \cdot 7\Omega$ and $V_{BC} = I \cdot 3\Omega$
- And $I = 1.2 \text{ A}$
- So $V_{AB} = 8.4 \text{ v}$ and $V_{BC} = 3.6 \text{ v}$





MODULE 3: ELECTRICITY & CIRCUITS ANALYSIS

UNIT 2: Kirchhoff's laws

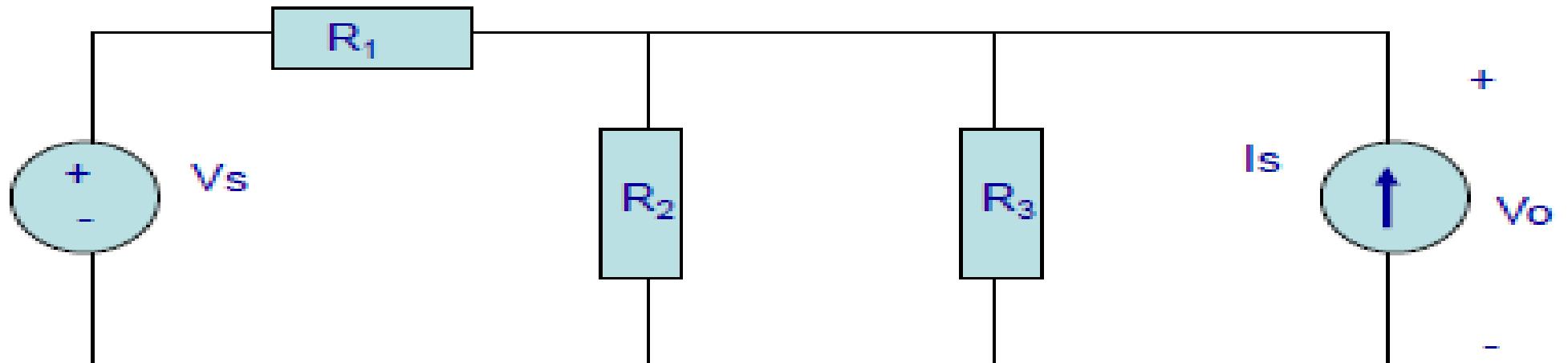


Circuit Definitions

- **Junction** – a junction is a point in a circuit where two or more conductors meet. It is also called a node or a branch point
- **Branch** – It represents a single element in a circuit such as the resistor, current source, voltage source etc.
- **Loop** – a collection of branches that form a closed conducting path.

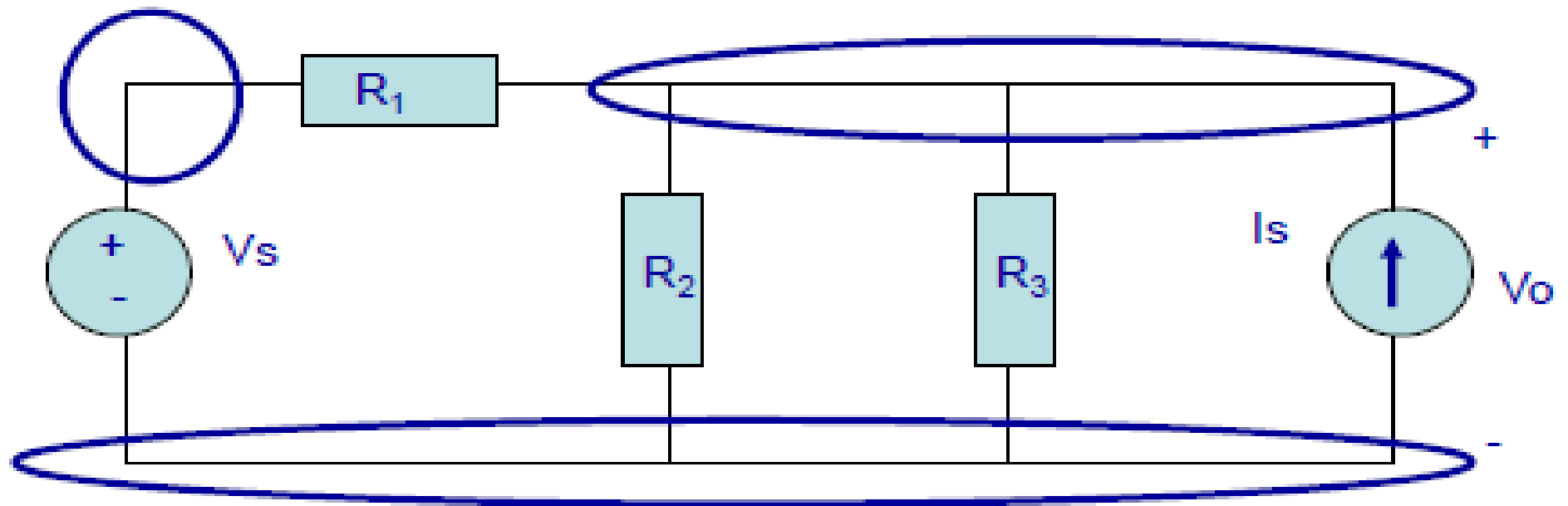
Example

- How many junctions, branches & loops?



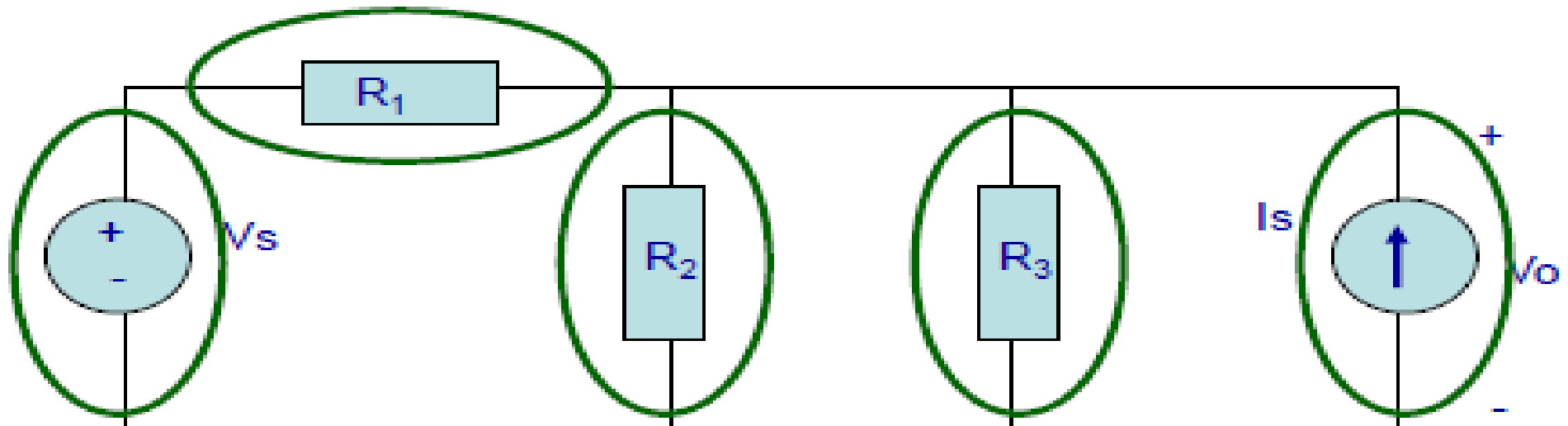
Example

- Three junctions



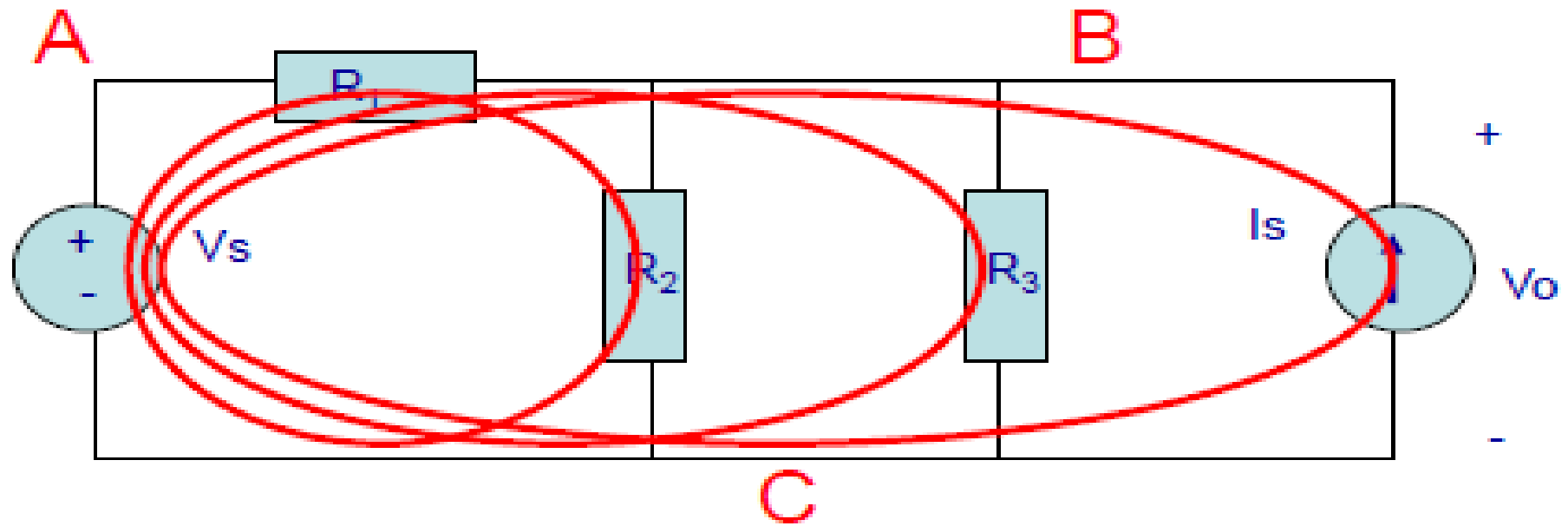
Example

- 5 Branches



Example

- Three Loops, if starting at node A



Kirchhoff's Voltage Law (KVL)

- The algebraic sum of the potential differences across all the elements around any closed circuit or loop must be zero.

A statement of Conservation of Energy

- $\Sigma V = 0$
- $\Sigma \text{ voltage drops} - \Sigma \text{ voltage rises} = 0$
- Or $\Sigma \text{ voltage drops} = \Sigma \text{ voltage rises}$



Conventions for Loop Rule

- The following conventions apply for determining the sign of ΔV across each circuit element.
- The travel direction is the direction that we choose to proceed around the loop. It may be **clockwise** or **anti-clockwise**.

Procedure for Applying Rules

travel direction → higher V lower V a b $\Delta V = V_b - V_a = -IR$	travel direction → lower V higher V a b $\Delta V = V_b - V_a = +IR$
travel direction → lower V higher V a b $\Delta V = V_b - V_a = +\mathcal{E}$	travel direction → higher V lower V a b $\Delta V = V_b - V_a = -\mathcal{E}$

Procedure for Applying Rules

- When traveling through a source in the direction from $-ve$ to $+ve$, the emf is considered to be $+ve$.
- When traveling through a source from $+ve$ to $-ve$, the emf source is $-ve$.
- When travel through a resistor in the same direction as the assumed current, the IR term is $-ve$.

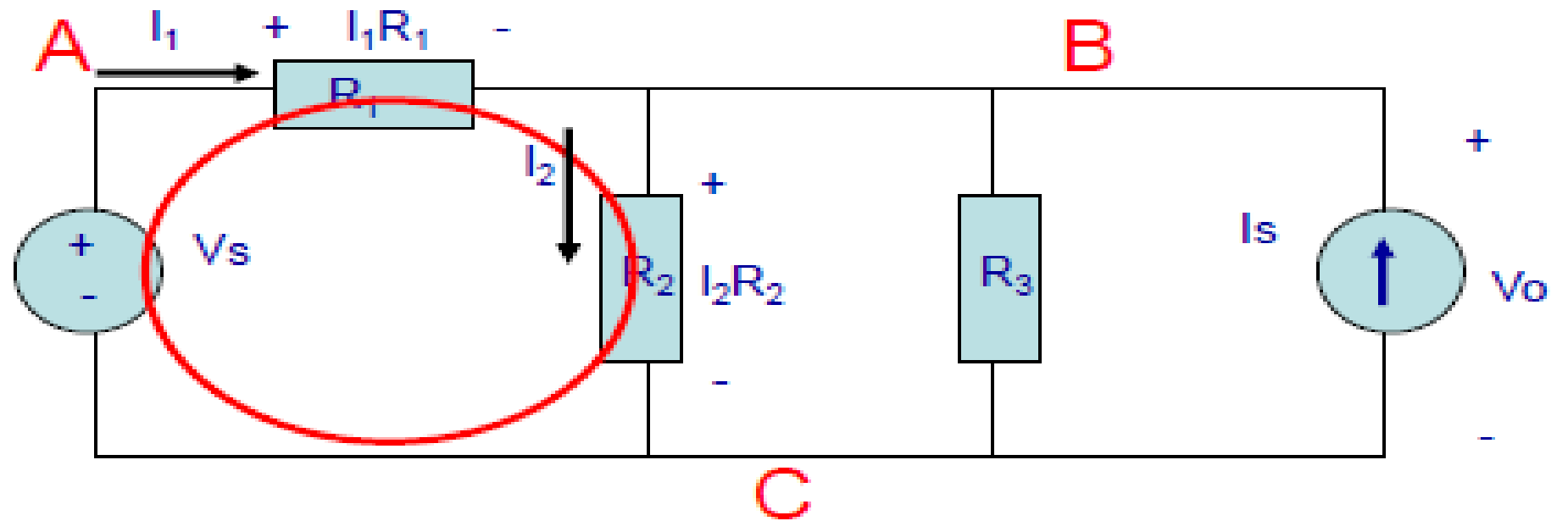


Procedure for Applying Rules Cont'd

- When we travel through a resistor in the direction opposite to the assumed current, the IR term is +ve.

Example

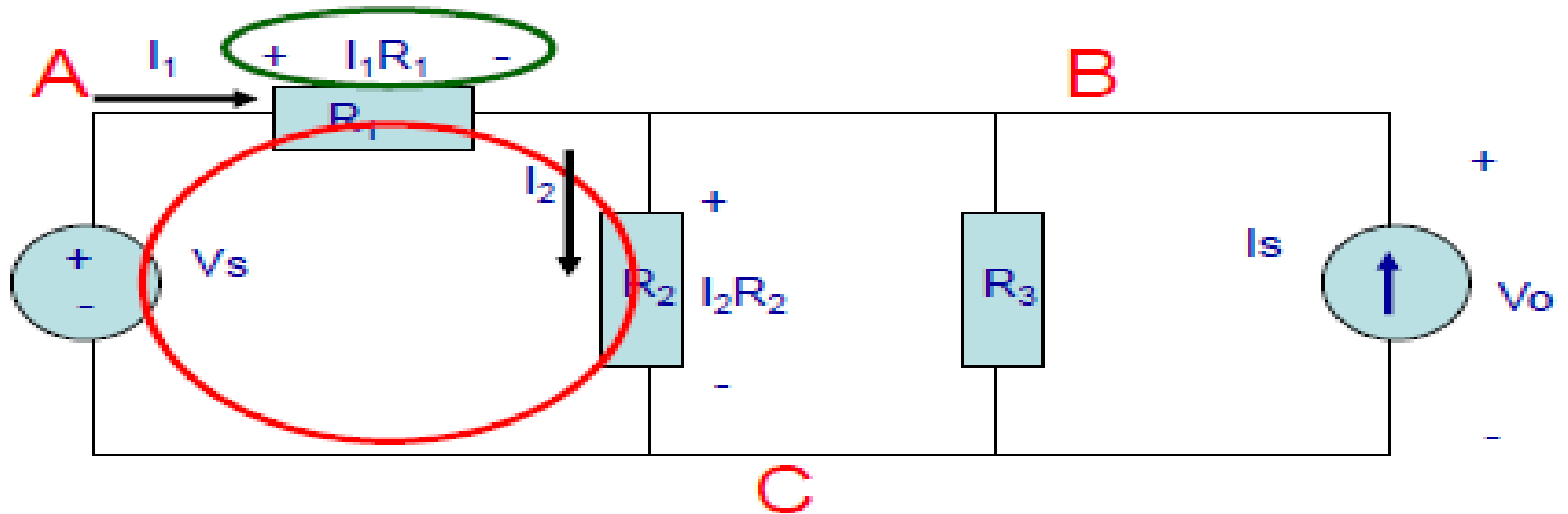
- Kirchhoff's Voltage Law around 1st Loop



Let us traverse the first loop in a clockwise direction

Example

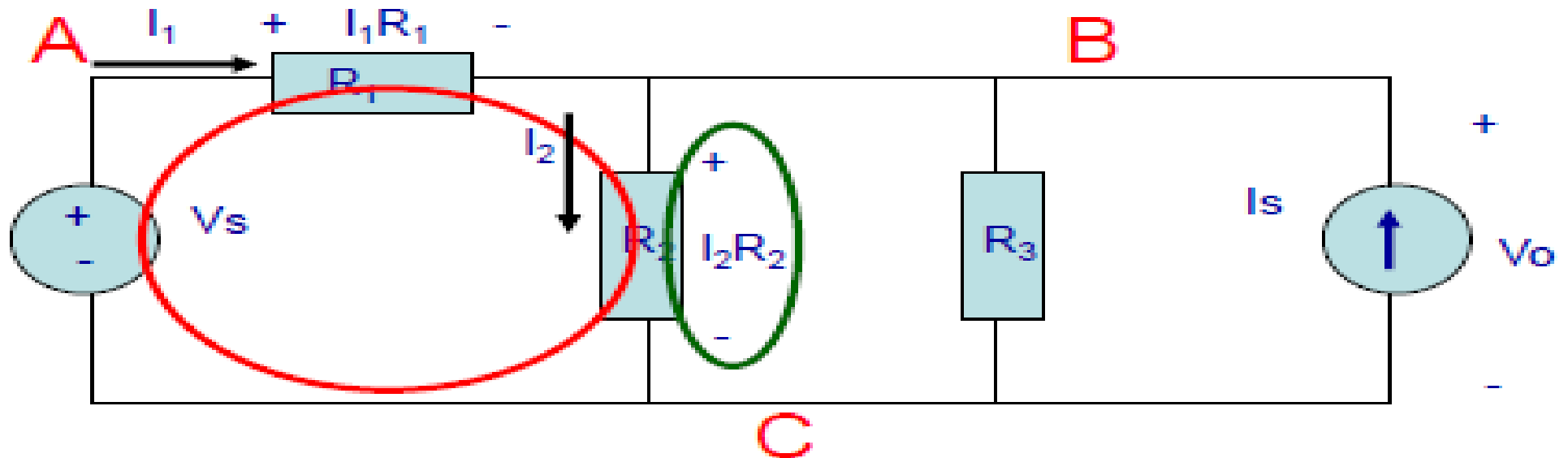
- Kirchhoff's Voltage Law around 1st Loop



Starting at node A, add the 1st voltage drop: $- I_1 R_1$

Example

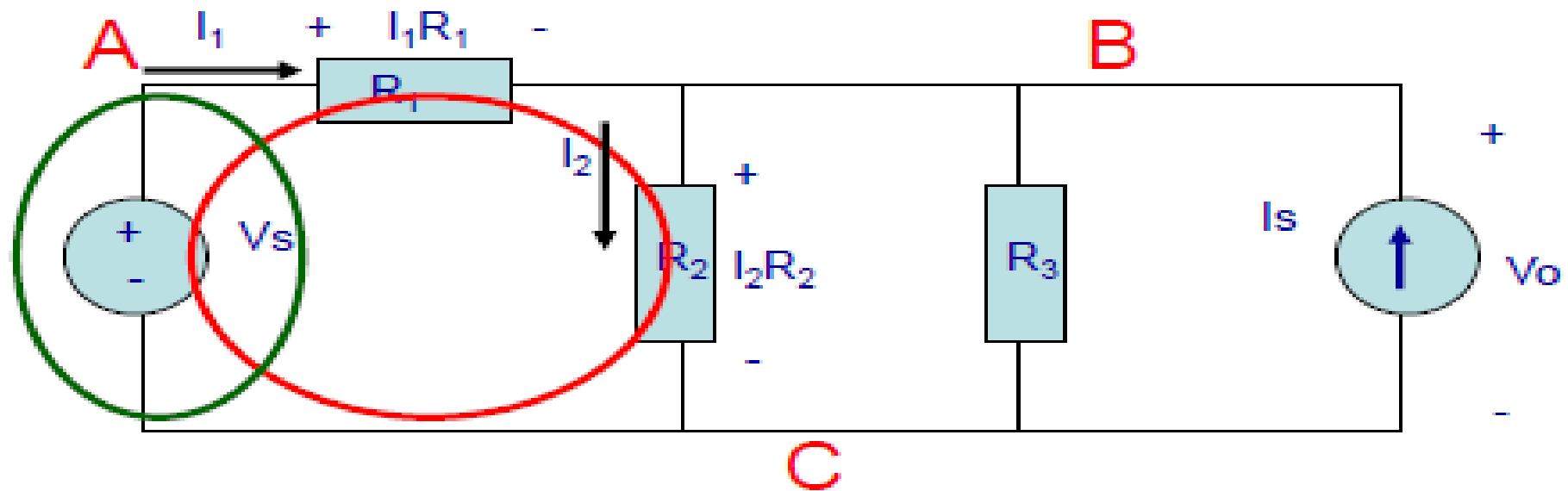
- Kirchhoff's Voltage Law around 1st Loop



Add the voltage drop from B to C through R_2 : $-I_1 R_1 - I_2 R_2$

Example

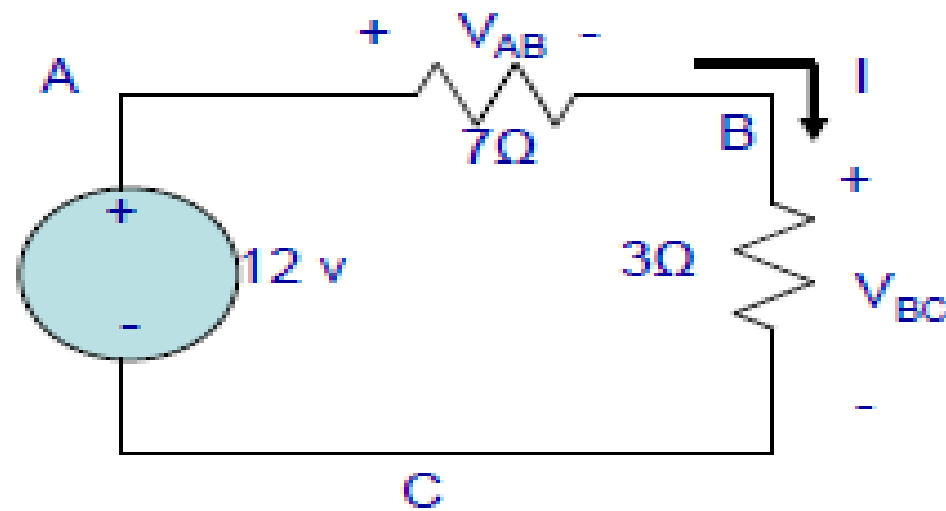
- Kirchhoff's Voltage Law around 1st Loop



Subtract the voltage rise from C to A through V_s : $-I_1R_1 - I_2R_2 + V_s = 0$

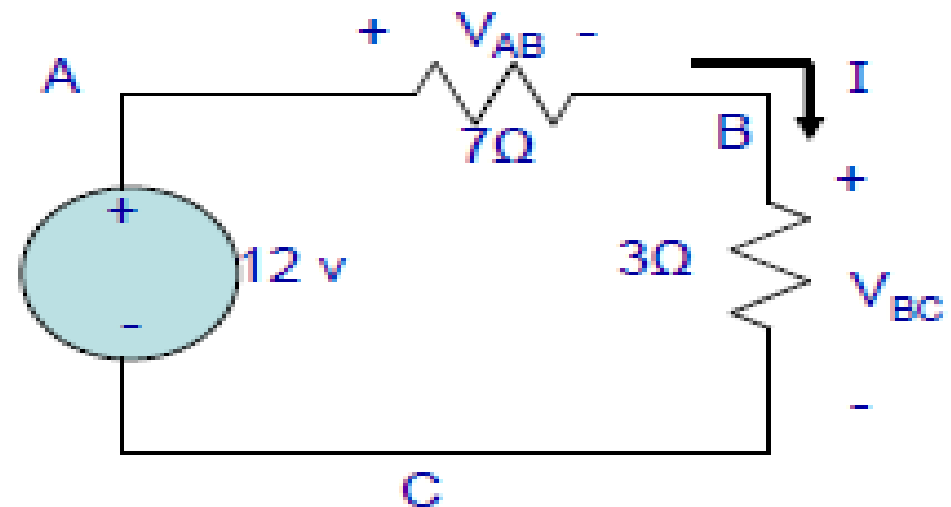
Circuit Analysis

- When given a circuit with sources and resistors having fixed values, you can use Kirchhoff's two laws and Ohm's law to determine all branch voltages and currents



Circuit Analysis

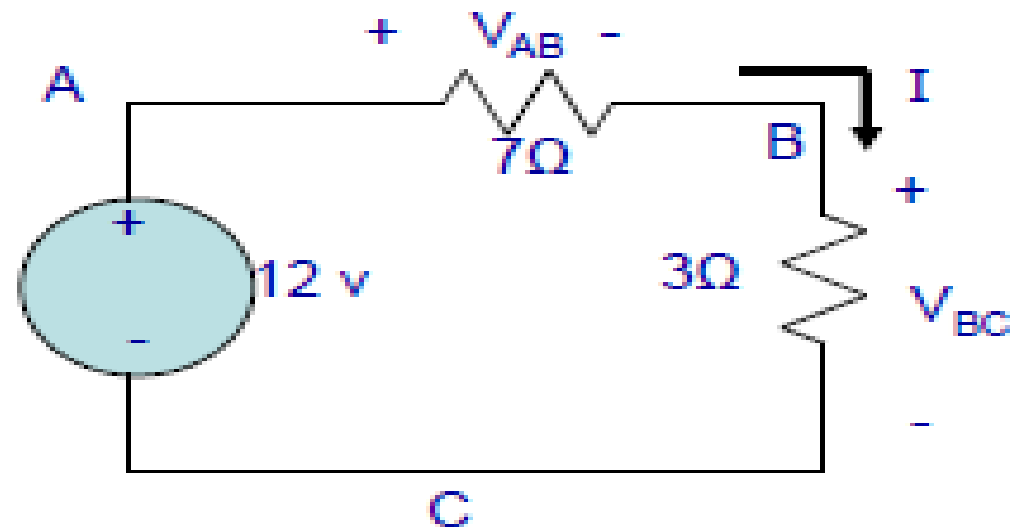
- By Ohm's law: $V_{AB} = I \cdot 7\Omega$ and $V_{BC} = I \cdot 3\Omega$
- Assume anti-clockwise direction



- By KVL: $V_{AB} + V_{BC} - 12 v = 0$
- Substituting: $I*7 + I*3 - 12 v = 0$
- Solving: $I = 1.2 \text{ A}$

Circuit Analysis

- Since $V_{AB} = I \cdot 7\Omega$ and $V_{BC} = I \cdot 3\Omega$
- And $I = 1.2 \text{ A}$
- So $V_{AB} = 8.4 \text{ v}$ and $V_{BC} = 3.6 \text{ v}$



Kirchhoff's Current Law (KCL)

- The algebraic sum of the currents entering any junction must be equal to the sum of the currents leaving that junction

- A statement of Conservation of Charge

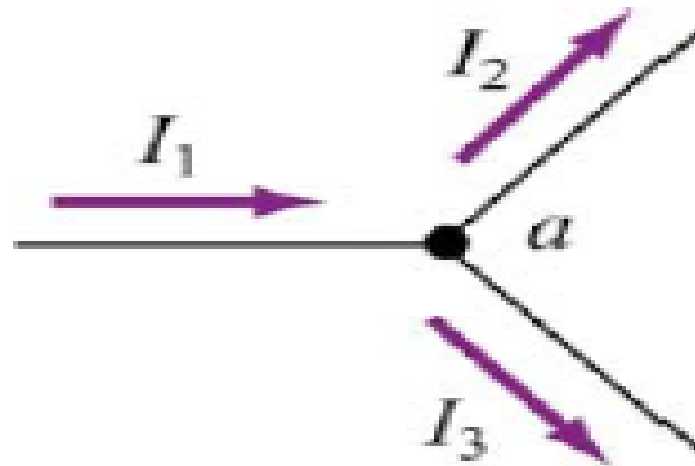
$$\sum I = 0$$

- $\sum \text{currents in} - \sum \text{currents out} = 0$

- Or $\sum \text{currents in} = \sum \text{currents out}$

Example

Express the currents in junction “a” as an equality.

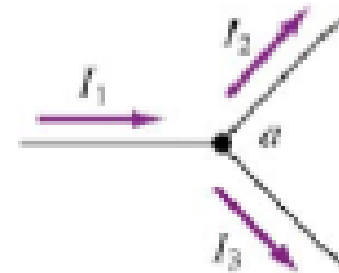


- Answer: Applying the junction rule, we have that:
- $I_1 = I_2 + I_3$

Example

If the currents exiting from junction “a” are to be of 2 amps each, what is the value for the current entering the junction?

- Recall the junction rule for this case:

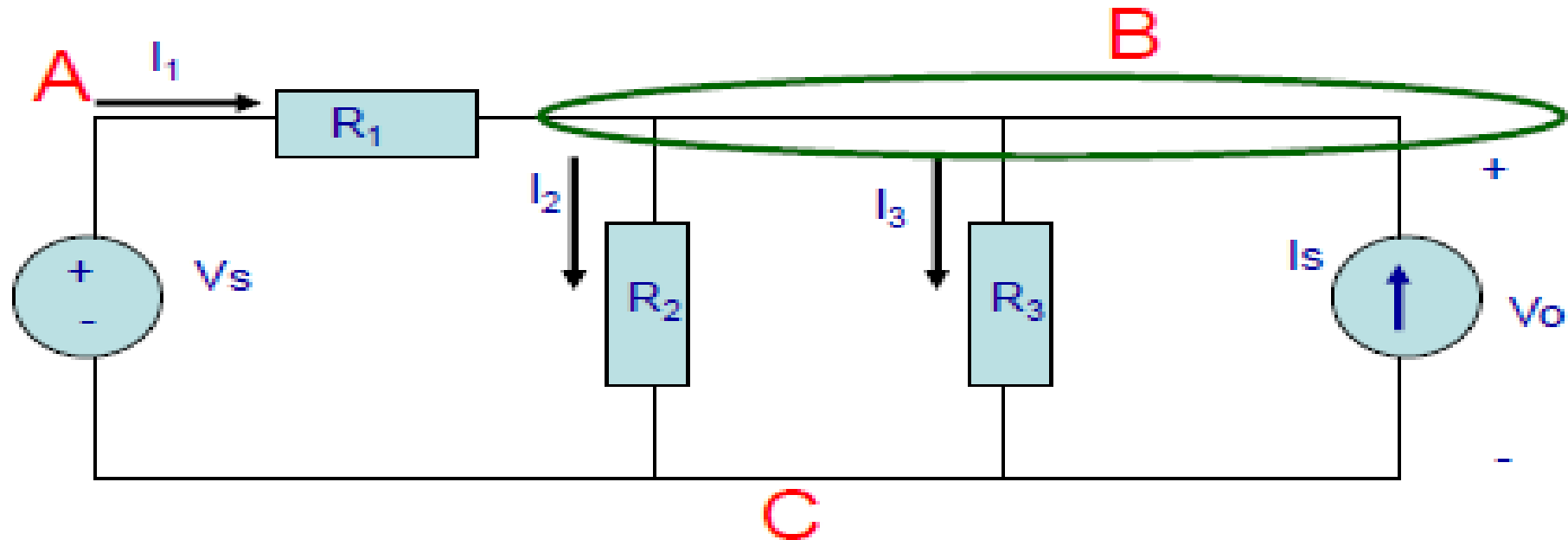


We know the following values:

- Then, we can solve for current entering the junction: $I_1 = I_2 + I_3$ and $I_2 = I_3 = 2 \text{ amps}$
- $I_1 = 2 + 2 = 4 \text{ amps}$

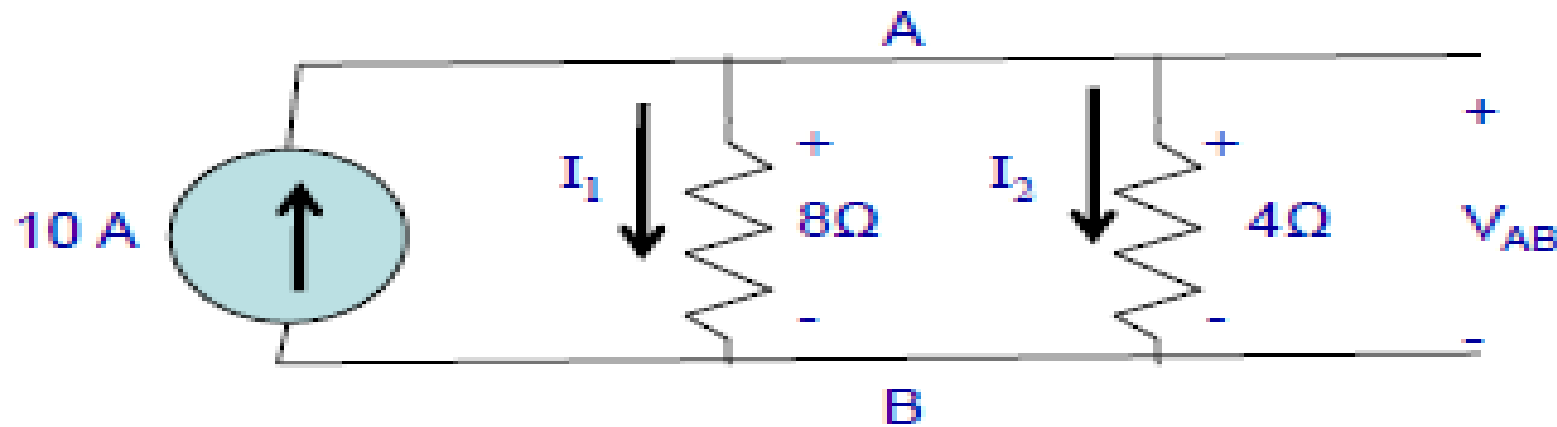
Example

- Kirchhoff's Current Law at B



The total current flowing in the circuit is given by $I_1 - I_2 - I_3 + I_s = 0$

Circuit Analysis



By KVL: $-I_1 \cdot 8\Omega = 0$ (a)

$-I_2 \cdot 4\Omega + 8 \cdot I_1 = 0$ (b)

Solving: $I_2 = 2 \cdot I_1$

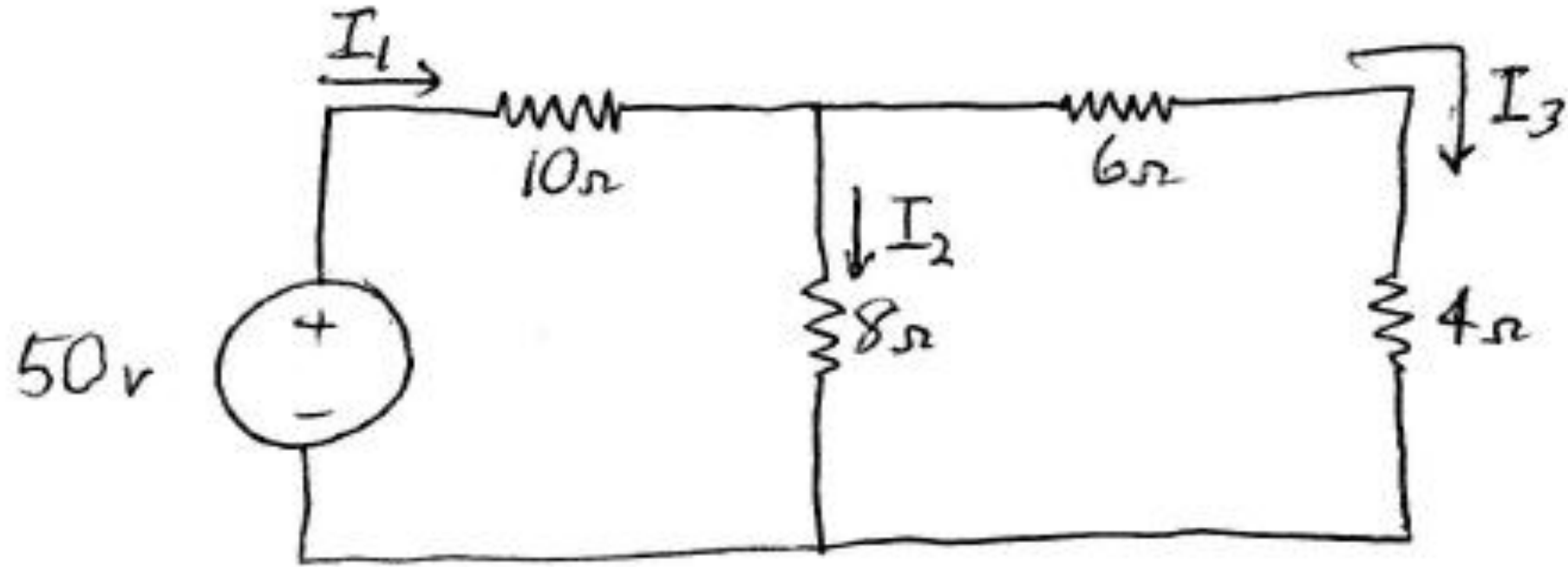
By KCL: $10A = I_1 + I_2$

Substituting: $10A = I_1 + 2 \cdot I_1 = 3 \cdot I_1$

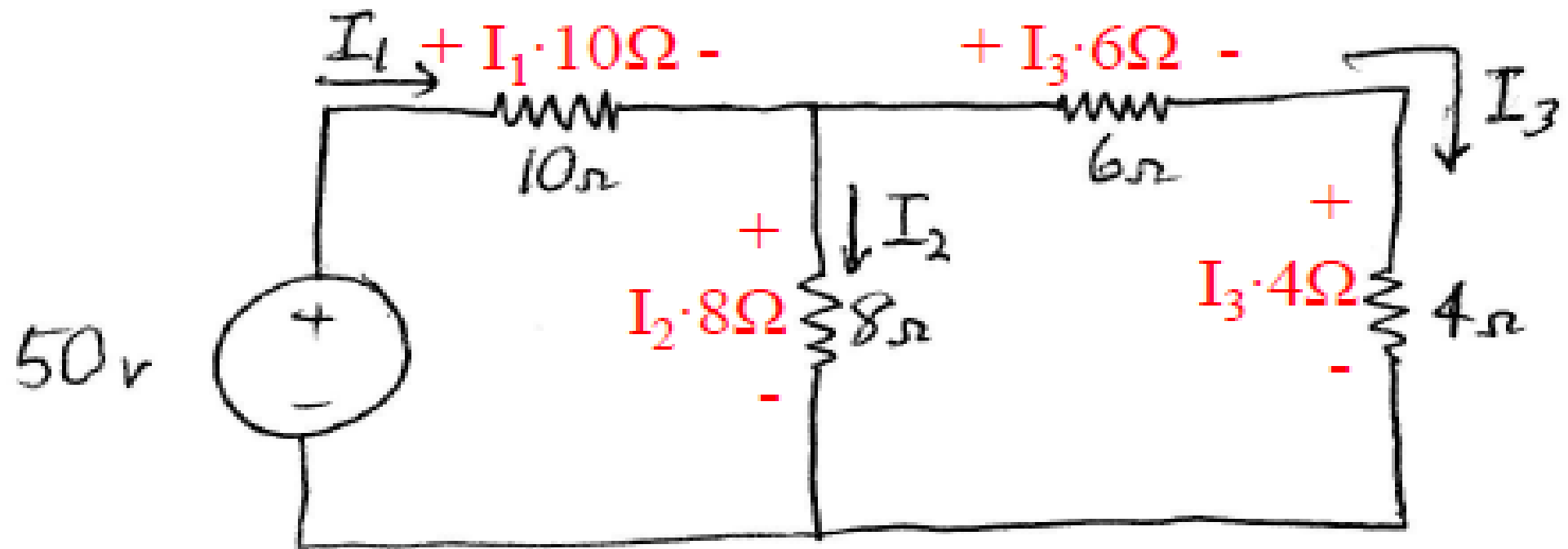
So $I_1 = 3.33 \text{ A}$ and $I_2 = 6.67 \text{ A}$

And $V_{AB} = 26.67 \text{ volts}$

Example Circuit

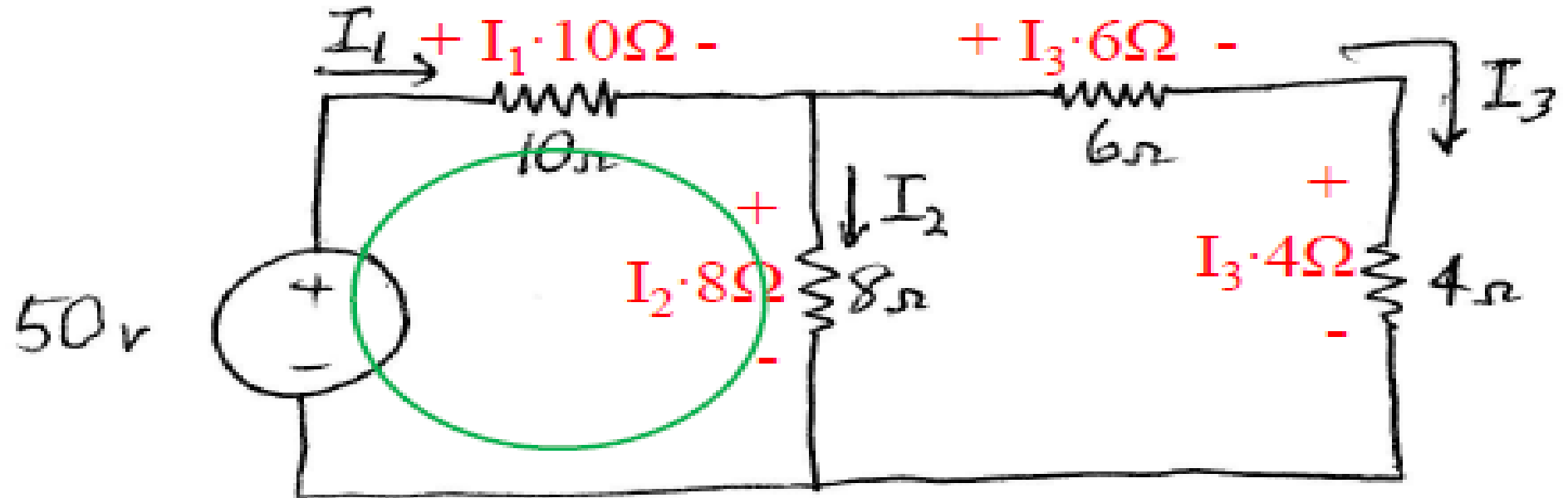


Solve for the currents through each resistor and the voltages across each resistor?
Assume **anti-clockwise** direction



Traversing the first loop in an anti-clockwise direction

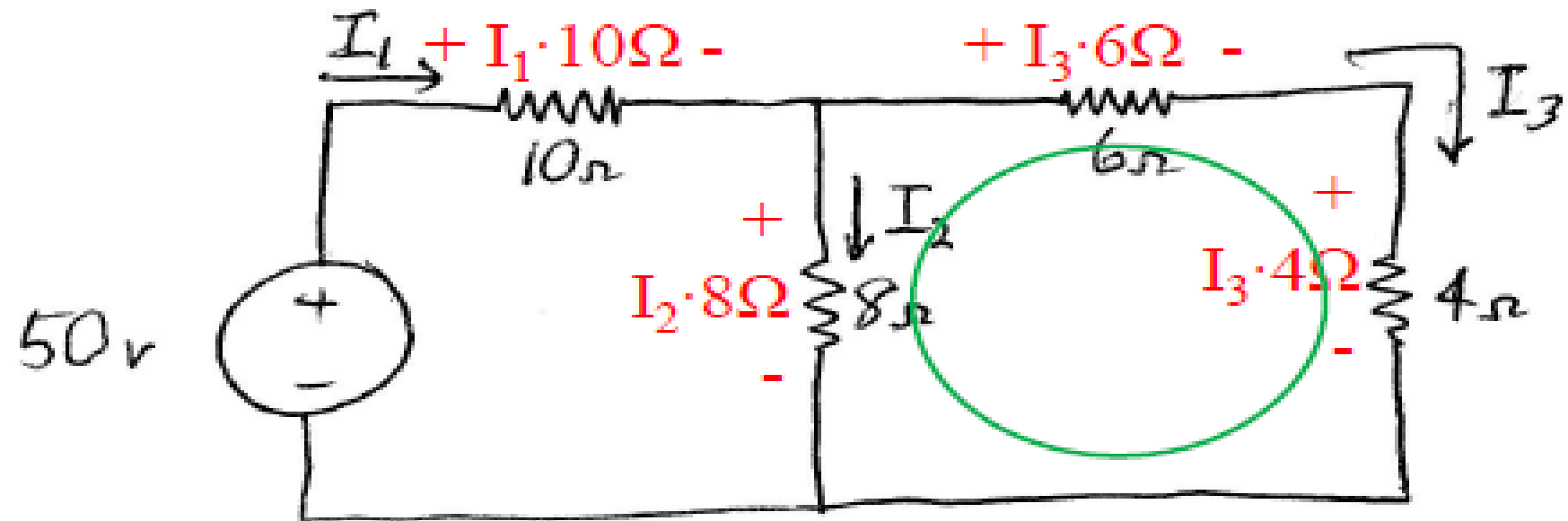
Example Circuit



Write 1st Kirchhoff's voltage law equation

$$-50 \text{ v} + I_1 \cdot 10\Omega + I_2 \cdot 8\Omega = 0 \quad \text{a}$$

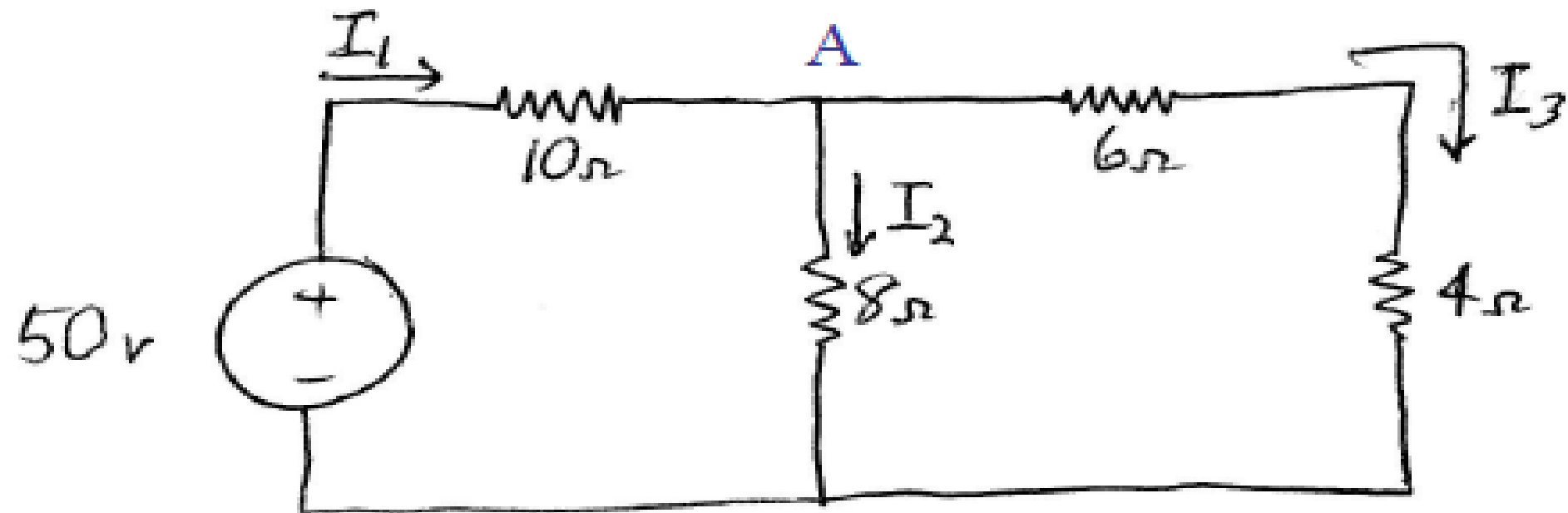
Example Circuit



Traverse the 2nd loop in an anti-clockwise direction
Write 2nd Kirchhoff's voltage law equation

$$-I_2 \cdot 8\Omega + I_3 \cdot 6\Omega + I_3 \cdot 4\Omega = 0$$

b



Write Kirchhoff's current law equation at A

$$I_1 = I_2 + I_3$$

c

Example Circuit

- We now have 3 equations and 3 unknowns, so we can solve for the currents through each resistor, that are used to find the voltage across each resistor

- Since $I_1 - I_2 - I_3 = 0$, $I_1 = I_2 + I_3$

- Substituting into the 1st KVL equation or a

$$-50 \text{ v} + (I_2 + I_3) \cdot 10\Omega + I_2 \cdot 8\Omega = 0$$

or

$$I_2 \cdot 18 \Omega + I_3 \cdot 10 \Omega = 50 \text{ volts}$$

Example Circuit

- But from the 2nd KVL equation i.e **b**, $I_2 = 1.25 \cdot I_3$
- Substituting into 1st KVL equation:

$$(1.25 \cdot I_3) \cdot 18 \, \Omega + I_3 \cdot 10 \, \Omega = 50$$

$$\text{Or: } I_3 \cdot 22.5 \, \Omega + I_3 \cdot 10 \, \Omega = 50$$

$$\text{Or: } I_3 \cdot 32.5 \, \Omega = 50$$

$$\text{Or: } I_3 = 50 \text{ volts} / 32.5 \, \Omega$$

$$\text{Or: } I_3 = 1.538 \text{ amps}$$

Example Circuit

- Since $I_3 = 1.538$ amps
 $I_2 = 1.25 \cdot I_3 = 1.923$ amps
- Since $I_1 = I_2 + I_3$,
- $I_1 = 3.461$ amps
- The voltages across the resistors:
 $I_1 \cdot 10\Omega = 34.61$ volts
 $I_2 \cdot 8\Omega = 15.38$ volts
 $I_3 \cdot 6\Omega = 9.23$ volts
 $I_3 \cdot 4\Omega = 6.15$ volts

Assume that current flows in clockwise directions use Kirchhoff's theorem to determine I_1 , I_2 and the p.d. across R_1 and R_2 ?

